

PHYSICS GRADE 10: INTRODUCTION TO ELECTRONICS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.3: Introduction to Electronics
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Conductors, insulators, semiconductors and superconductors – Intrinsic semiconductors – Extrinsic semiconductors (n-type and p-type) and doping – Effect of temperature on the conductivity of materials – Superconductors and their applications (e.g. MRI magnets) – Applications of semiconductors and superconductors in solar panels, microchips and ICT devices
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - classify materials as conductors, insulators, semiconductors and superconductors based on their electrical properties, - describe intrinsic and extrinsic (n-type and p-type) semiconductors, - explain how doping and temperature change the conductivity of semiconductors, - explain the special properties and uses of superconductors, - relate the properties of these materials to their applications in household wiring, solar panels, microchips and MRI machines, - appreciate the role of electronics in Kenya's technological growth and renewable energy.
Core Competencies	– Critical thinking and problem solving: the learner compares materials, analyses evidence on conductivity and temperature, and explains why each material suits its application. – Communication and collaboration: the learner works in groups to test materials, discuss findings and present explanations of the energy band and doping models. – Digital literacy: the learner uses the PhET simulation to investigate semiconductors and doping. – Learning to learn: the learner revises a cumulative model of conduction as new evidence is gathered across the lessons.
Core Values	– Responsibility: the learner handles electrical materials, wires and any liquid nitrogen demonstration safely and uses electrical devices responsibly. – Unity: the learner cooperates in groups to carry out investigations and build shared models. – Integrity: the learner reports observations honestly, even when they differ from predictions.
Science & Engineering Practices	– Asking questions and defining problems: students generate questions for the Driving Question Board about why materials differ in conductivity. – Developing and using models: students build and revise particle and energy-band models of conductors, insulators,

	semiconductors and superconductors. – Planning and carrying out investigations: students test copper, glass, wood, plastic and a diode for conduction. – Analyzing and interpreting data: students interpret conductivity-temperature trends. – Constructing explanations: students explain household wiring, solar cells and MRI magnets using evidence. – Obtaining, evaluating and communicating information: students analyse case studies of solar panels, microchips and superconducting magnets.
Pertinent & Contemporary Issues (PCIs)	– Safety and security: safe handling of electrical wiring and insulation to prevent shocks and fires in homes and workshops. – Environmental education: use of solar energy as a clean renewable energy source. – Education for sustainable development: promoting renewable energy and proper disposal of electronic waste (e-waste).
Career Connections	– Electrical technician or electrician: choosing conductors and insulators for safe wiring in homes. – Solar installation technician (e.g. in Kisumu, Kitui or Garissa): installing and maintaining solar PV systems. – Electronics and ICT engineer: designing circuits and microchips for phones and computers. – Biomedical engineer or radiographer: operating MRI machines at hospitals such as Kenyatta National Hospital. – Materials scientist or physicist: researching new semiconductors and superconductors.
Focus for Lessons	This unit helps learners explain why materials differ in how they carry electric current and how this is used in wiring, solar panels, microchips and MRI machines. Learners predict, test and model conduction, build up from conductors and insulators to semiconductors, doping and superconductors, and finally apply their ideas to Kenyan renewable energy and technology cases.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? KICD KEY INQUIRY QUESTIONS: 1. How do semiconductors and superconductors differ from ordinary conductors and insulators? 2. How is electronics applied in day-to-day life?
Anchoring Phenomenon	In a Kenyan home in Nairobi, the electrician wires a house with copper cable covered in plastic. Copper carries electricity efficiently, while the plastic covering, wood and glass do not, so a person can safely hold the insulated wire while it carries current. Students test copper wire, glass, wood, plastic and a diode chip in a simple circuit and see a bulb light only for some of them, with the diode conducting in one direction only. It is puzzling because all these materials are made of atoms with electrons, yet they behave so differently. Why do metals like copper carry current easily while plastic and wood do not, and what makes the semiconductor in between so useful?
Storyline Thread	Lesson 1: Students predict which materials (copper, glass, wood, plastic, a diode) will light a bulb, test them, and start the Driving Question Board and a first model of conduction. Lesson 2: Students classify conductors, insulators and semiconductors, link the classification to household wiring safety and the role of insulators, and revise the model. Lesson 3: Students investigate intrinsic semiconductors (silicon, germanium) and explain why they conduct a little and how temperature increases their conductivity. Lesson 4: Students explain doping to make extrinsic n-type and p-type semiconductors and why doping makes semiconductors useful in electronics. Lesson 5: Students use the PhET simulation to explore conductivity, doping and energy bands, and test and refine their models with digital evidence.

	Lesson 6: Students examine superconductors, critical temperature and zero resistance (with a liquid nitrogen demonstration or video if possible), and explain superconducting MRI magnets. Lesson 7: Students analyse case studies of solar panels, microchips and superconducting magnets in Kenya's technological growth, finalise their models and answer the Driving Question.
Supporting Phenomena	– Semiconductors in solar cells and microchips control the flow of electricity, e.g. in solar panels on rooftops in Kenya and in mobile phones. – Adding small amounts of impurities (doping) changes the conductivity of semiconductors. – Conductors become less efficient at higher temperatures, semiconductors become more conductive, and superconductors lose all resistance below a critical temperature. – Superconducting magnets in MRI machines produce strong magnetic fields without energy loss. – Plastic, rubber and glass insulators prevent unwanted current flow and protect users.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (one double or one single lesson; if single, run Observe and Explain in 25 minutes and keep DQB and Model to 15 minutes)): Which Materials Light the Bulb? Launching the Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon: Mr Otieno, an electrician in Nairobi, wires a house with copper cable covered in PVC plastic and safely holds the insulated wire while it carries current. Learners predict, test and compare materials in a simple circuit, discover that only some materials let current flow, and begin to ask why.
Knowledge	Materials differ in whether they let current flow. In the test circuit, copper lights the bulb brightly, while glass, dry wood and plastic do not. A silicon diode lights the bulb in one direction only (forward-biased: lit; reverse-biased: off). So 1 of 4 ordinary materials conducts. All the materials are made of atoms with electrons, which raises the puzzle of why they behave so differently.
Skills	Making and justifying predictions (NGSS SEP: Asking Questions and Defining Problems); planning and carrying out a fair test in a circuit with a fixed test order (SEP: Planning and Carrying Out Investigations); recording and organising observations in a table (SEP: Analyzing and Interpreting Data); drawing a first model of electrons in materials (SEP: Developing and Using Models); communication and collaboration in groups.
Attitudes	Curiosity about why materials behave differently; respect for peers' ideas and open-mindedness when predictions turn out wrong; responsibility when handling dry cells, clips and wires; appreciation of the work of electricians such as Mr Otieno and of electronics in day-to-day life.
Key Inquiry Question	1. How do semiconductors and superconductors differ from ordinary conductors and insulators? 2. How is electronics applied in day-to-day life? (Launched today, answered across Lessons 2 to 7.)
Purpose in Storyline	Lesson 1 launches the phenomenon and the unit. It builds on nothing earlier. Learners gather first-hand evidence from the five-material test (copper, glass, dry wood, plastic, diode), start the Driving Question Board (DQB) with their own questions, and draw a first class model of electrons and atoms. Later lessons revise this model: Lesson 2 classifies materials by resistivity, Lesson 3 explains pure semiconductors, Lesson 4 doping, Lesson 5 tests band models with a simulation, Lesson 6 superconductors and MRI, Lesson 7 the final answer.
Safety Notes	Use only two 1.5 V dry cells in series (3 V) and a 3 V torch bulb; never connect to mains (240 V) sockets. Do not short-circuit the cells with copper wire: keep the copper in the gap only briefly (a few seconds) and disconnect after each reading because the wire and clips may warm. Handle the glass piece with care (use a smooth-edged piece or rod; report any chips). Connect the 1N4007 diode with correct orientation and do not leave it connected for long. Keep liquids and water away from the circuit. Wash hands after handling cable insulation offcuts. Teacher must check every circuit before learners close it.

B. LESSON OVERVIEW

Ms Achieng opens with the image of Mr Otieno wiring a Nairobi house with copper cable covered in PVC plastic, holding the insulated wire while it carries current at 240 V. She asks learners what makes that safe. Learners predict which of five materials (copper, glass, dry wood, plastic, diode) will light a 3 V bulb in a simple circuit and explain why. In groups, like Wanjiku and Kamau, they test the materials in the fixed order and record results: copper lights the bulb brightly; glass, dry wood and plastic do not; the diode lights the bulb forward-biased and not reverse-biased. In the Explain phase, learners compare predictions to evidence, note that only 1 of 4 ordinary materials conducts, and that the diode behaves differently from all of them. They name the puzzle: all materials are made of atoms with electrons, yet behave so differently. They then write questions for the Driving Question Board (DQB) and draw a first class model of electrons in a conductor and in an insulator, which will be

revised in every later lesson. Lesson 1 does not explain energy gaps or resistivity; those come later.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look at a picture or the teacher's description of Mr Otieno, an electrician in Nairobi, wiring a house with copper cable covered in PVC plastic and holding the insulated wire safely while it carries current. Individually, then in pairs, they complete a prediction table for the five materials in the fixed order (copper wire, glass, dry wood, plastic strip, silicon diode): Will the bulb light? Yes / No / Only one way. Beside each prediction they write one reason ('I think this because...'). For the diode they sketch how they might connect it and predict whether direction matters. Pairs then share predictions with a neighbouring pair.	Show the image and say: 'Mr Otieno is holding a wire carrying 240 V and he is not hurt. What do you notice about the wire?' WAIT TIME 10 seconds before taking answers. Say: 'Today we will test materials in a very safe 3 V circuit. Before we test, I want your prediction and your reason. There are no wrong predictions today.' Hold up each material in the fixed order. Give 3 minutes for silent individual prediction, then 3 minutes pair talk. Cold-call 3 pairs: 'What did you predict for wood, and what is your reason?' Then ask: 'Who predicted differently?' and cold-call 2 more. Record the class tally on the board (for each material: number predicting yes, no, one way) without saying who is right. Ask: 'What do you think happens with the diode if we flip it around?' and WAIT TIME 10 seconds.	Predict-Observe-Explain (POE) with a public class tally. Committing to a prediction and a reason, in writing, exposes learners' initial ideas (for example 'wood is natural so it conducts' or 'all metals and all plastics behave alike'). Because the tally is recorded before testing, learners later feel the surprise or confirmation of the evidence, which motivates the need for an explanation.	Check that 100% of learners have written a prediction and a reason for all five materials (walk round with a checklist). Listen for reasoning types: appeals to appearance ('shiny'), to everyday use ('wires are covered in plastic'), or to particles ('electrons move'). Note at least 5 learners whose reasons mention a particle idea or everyday safe use of wire and whose ideas you can call on later. Record the class tally as baseline data.
Observe Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators	In groups of 4 (roles: circuit builder, tester, recorder, reporter), learners build the test circuit: two 1.5 V dry cells in series (3 V), a 3 V torch bulb and crocodile clips, leaving a gap in the circuit. First they close the gap with a clip-to-clip touch to check that the bulb works. Then they test the materials in the fixed order: (1) copper wire 10 cm, (2) glass rod/piece, (3) dry wood (ordinary ruler/stick), (4) plastic strip (stripped cable insulation), (5) silicon diode	Demonstrate the safety rules first: 'Cells and bulb only. Never touch these clips to a socket. Disconnect after each reading.' Check each circuit before groups test anything: 'Hands up when your bulb lights with the clips touching, and I will come to check.' Circulate and ask: 'What does the bulb tell you about the current?' and 'What would you expect to see if the current were very small?' Remind groups to keep to the fixed order. When groups reach the diode say: 'Test	Structured data collection with a shared class results table. A fixed test order and a common circuit let all groups produce comparable evidence, and pooling results lets learners see patterns (1 of 4 ordinary materials conducts) and an anomaly (the diode conducts one way only). Highlighting the diode as an anomaly generates real questions rather than teacher-given ones.	Observe whether groups: (a) build a working circuit and check it with touching clips, (b) test in the fixed order, (c) record bright/off for each material and both diode directions accurately, (d) disconnect after each reading. Collect recorded tables: at least 80% of groups should record copper bright, glass off, wood off, plastic off, diode lit one way and off the other. Note groups who record the diode result in words such as 'works only one way' or who spontaneously ask

Practice Source: CK-12 Search ARES for similar readings	(1N4007), first forward-connected, then reversed. They record in a table whether the bulb is bright, off or lit, and also note anything else, such as the wire feeling warm. Like Wanjiku and Kamau, they compare results with their predictions. Expected results: copper, bulb bright; glass, off; wood, off; plastic, off; diode forward-connected, lit; diode reverse-connected, off.	it, then reverse it. Tell me exactly what you see each time.' WAIT TIME 10 seconds after posing 'Why might the diode behave differently from the copper?' without answering it. Cold-call 4 reporters to give one result each in the fixed order and write the class results on the board. Resolve any odd result (for example a dull bulb with a flat cell) by asking the group to retest rather than dismissing it.		why.
Explain Phase  VIDEO: 03 Intrinsic And Extrinsic Semi Conductors <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12 Search ARES for similar readings	Learners compare the class results with the prediction tally and discuss in groups: Which predictions were correct? Which surprised us? They state the pattern in a claim-evidence-reasoning frame: 'Claim: only copper lets current flow easily. Evidence: the bulb lit brightly with copper but stayed off with glass, wood and plastic.' They note that 1 of 4 ordinary materials conducts, and that the diode conducts in one direction only (forward-biased: lit; reverse-biased: off). They link back to Mr Otieno: the copper core carries current and the plastic (PVC) cover does not, which is why he can hold the insulated wire. Then they name the puzzle: all of these materials are made of atoms with electrons, yet they behave so differently. In pairs they write one 'I wonder' about why. Learners use the words conductor and insulator for the first four materials, and the term diode for the fifth, but not yet explain why.	Say: 'Look at the board. Compare our results with our predictions. What stands out?' WAIT TIME 15 seconds. Cold-call 3 learners for a pattern, then 2 more for a surprise. Push for evidence: 'What did you see that tells you that?' and 'How do you know the bulb is off because of the material, and not a flat cell?' (point to the touching-clips check). Introduce the vocabulary: 'We call materials that let current flow easily conductors, and materials that almost do not let current flow insulators. We say almost none, not zero.' Return to the phenomenon: 'Mr Otieno holds the insulated wire. Which of our materials is the copper and which is the cover? What does our test say about why this is safe?' Cold-call 2 learners. Then pose the puzzle: 'Everything we tested is made of atoms, and atoms have electrons. So why do copper and plastic behave so differently? And what is going on with the diode, which seems to be in between?' WAIT TIME 15 seconds, then let pairs talk for 2 minutes. Do NOT give the energy-gap or resistivity explanation; say 'That is exactly what we will find out in the coming lessons.'	Claim-Evidence-Reasoning (CER) with explicit comparison of predictions to data. Writing claims tied to specific bulb observations stops learners from relying on appearance or assumption. Naming the puzzle (same building blocks, different behaviour) creates a need to know that drives the unit, and the diode anomaly signals that the conductor/insulator split is not the whole story.	Listen for claims that cite specific evidence (for example 'copper bright, plastic off'). Check that learners distinguish conductor/insulator vocabulary from observation and that they use 'almost no current' rather than 'no electricity at all'. Look for at least 6 'I wonder' statements that go beyond 'what is the diode?' to particle ideas (for example 'do electrons move in copper but not in plastic?'). Note any misconception, such as 'plastic has no electrons' or 'wood is a conductor because it is natural', and record it for the DQB and for later lessons.

Driving Question Board (DQB) Creation  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice Source: CK-12  Search ARES for similar readings	Learners write their questions on sticky notes or paper strips (one question per note) and post them on the class DQB under the heading of the Driving Question: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? Questions come from their predictions, results and 'I wonder' statements. In groups they cluster questions into categories (for example: why copper conducts; why plastic, glass and wood do not; what the diode is and why it works one way; how this links to solar panels, phones and computers; what MRI machines have to do with it). Each group reads out one question it most wants answered. Learners also copy the two KICD key inquiry questions onto the DQB.	Post the Driving Question and the two key inquiry questions at the top of the board. Say: 'Write every question you have, one per note. Good questions begin with why or how.' Give 4 minutes. Model one: 'Why can Mr Otieno touch plastic-covered wire but not bare copper?' Collect and cluster with learners: 'Who has a question that is like this one? Come and place it here.' Cold-call 4 learners to read and place a question. Ask: 'Which question could we answer by testing something? Which need us to learn more about what is inside the materials?' WAIT TIME 10 seconds. Tell learners: 'We will return to this board in every lesson and move notes to Answered as our evidence grows.' Ensure questions about solar panels, computers, M-Pesa phones and MRI machines at Kenyatta National Hospital are on the board, adding one yourself if none appears.	Driving Question Board (DQB) as a public record of learners' own questions. Clustering questions shows learners how their wonderings connect to one big question, gives them ownership of the unit, and creates a visible tracker to return to for questions answered, revised or new in Lessons 2 to 7.	Count questions: aim for at least 2 per group, with at least 3 questions in the whole class about why the materials differ at the particle level and at least 2 about applications (solar panels, phones, MRI). Check that clusters are sensible and that quieter learners have contributed at least one note. Record frequent question types to plan which later lessons to emphasise.
Model Building Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations	On A4 paper or in their exercise books, learners draw a first model to explain why copper lets current flow but plastic (the PVC cover) does not. They draw a small piece of copper and a small piece of plastic, show atoms and electrons in each in any way they choose, and use arrows or labels to show where electrons move or do not move when the 3 V cells are connected. They add a third box for the diode with a question mark and a note: 'one way only – we do not know why yet'. They write a one-sentence 'what my model says' and one 'what my model cannot explain yet'. Groups combine their drawings on a large class model poster by agreeing on	Say: 'Use only what you have seen today. Draw what might be happening inside copper and inside plastic. It is fine to be unsure.' Circulate and ask: 'What does the arrow show? Why did you draw the electrons there?' and 'How does your model explain the bulb lighting with copper? How does it explain the bulb staying off with plastic?' Do not correct models; instead ask 'What evidence supports that?' Cold-call 3 groups to show their drawings, and ask the class: 'What do these models have in common? Where do they differ?' WAIT TIME 10 seconds. Build the class model poster from agreed features and write the disagreements as	Initial model construction and public comparison of models. Putting ideas about electrons and atoms into a drawing makes thinking visible, reveals alternative ideas, and gives learners something concrete to test and revise. Recording what the model cannot yet explain (the diode, the difference between copper and plastic) sets up the revision cycle across the unit.	Look for models that (a) show copper and plastic differently, (b) include electrons in both materials, (c) show some link between electron movement and the bulb lighting, and (d) acknowledge the diode as unexplained. Accept varied drawings. Note models that claim plastic has no electrons or that electrons are used up in the bulb; flag them to revisit, not to correct now. Check that each learner can state one thing the model explains and one thing it cannot.

(English) Search ARES for similar readings	common features and marking disagreements with question marks (for example, do all electrons move, or only some?). The class model is pinned beside the DQB.	question marks. Close by saying: 'This is our first class model. In each lesson we will test it and revise it with new evidence.' Link to home: 'At home, look at a cable or extension lead. Which part is copper and which is plastic? Bring a question to add to the DQB.'		
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D. TEACHER REFLECTION

After the lesson, review: (1) Did learners commit to written predictions with reasons before testing, and what were the most common ideas (appearance, use, particles)? (2) Did all groups obtain the fixed results (copper bright; glass, wood, plastic off; diode lit forward, off reverse)? If not, was it a flat cell, a loose clip or the wrong diode orientation? (3) Which learners used the word 'electrons' unprompted, and which held misconceptions such as 'plastic has no electrons'? (4) Did the DQB contain questions on both particle-level explanations and real applications (solar panels, phones, MRI at Kenyatta National Hospital)? (5) Did I resist explaining energy gaps and resistivity early, leaving them for later lessons? Use the DQB clusters and the first class model to decide which questions to emphasise in Lesson 2, where learners classify materials by resistivity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tested copper wire, glass, dry wood, plastic and a silicon diode in a 3 V circuit with a torch bulb. The bulb was bright with copper, and off with glass, dry wood and plastic. With the diode connected forward (forward-biased) the bulb lit, and with it reversed (reverse-biased) the bulb was off. So 1 of 4 ordinary materials conducts, and the diode conducts in one direction only.
What did I learn?	Materials differ in whether they let current flow: copper is a conductor, while glass, wood and plastic are insulators that let almost no current through, which is why Mr Otieno can safely hold plastic-covered copper cable. The diode is different from both groups because it depends on direction.
How does this explain the phenomenon?	We cannot yet explain why copper lets current flow and plastic does not, even though both are made of atoms with electrons, or why the diode works one way only. Our first class model shows electrons in each material, but it cannot explain these differences yet. Our DQB questions will guide what we investigate next.

LESSON 2 (80 minutes (2 x 40-minute periods)): Classifying Conductors, Insulators and Semiconductors

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use resistivity data to classify materials as conductors, insulators or semiconductors and explain why copper cable covered in PVC makes Kenyan household wiring both efficient and safe.
Knowledge	Materials are classified by resistivity (ρ , in ohm metre) at about 20 C. Order from lowest to highest: copper 1.7×10^{-8} , germanium 0.46, silicon (pure) 2.3×10^3 , dry wood about 10^8 to 10^{11} , glass about 10^{10} to 10^{14} , PVC about 10^{13} or more. Conductivity is $1/\rho$. Copper is a conductor; glass, wood and PVC are insulators that allow almost no current; silicon and germanium are semiconductors with resistivity in between. Silicon is about 1.4×10^{11} times more resistive than copper. In Mr Otieno's wiring (2.5 mm squared copper, PVC insulation, mains 240 V, 50 Hz) the copper carries current efficiently and the PVC keeps the current inside the wire so a person can hold the insulated cable safely.
Skills	Analysing and interpreting data (ordering resistivity values across many powers of ten); constructing classifications from evidence; using mathematics (powers of ten, ratios, $1/\rho$); developing and revising models; constructing explanations that link properties to safety.
Attitudes	Responsibility for electrical safety at home; curiosity about why materials behave so differently; respect for peers' ideas during discussion; appreciation of the skill of electricians such as Mr Otieno.
Key Inquiry Question	How do semiconductors and superconductors differ from ordinary conductors and insulators? (Focus: ordinary conductors, insulators and semiconductors.) How is electronics applied in day-to-day life? (Focus: household wiring.)
Purpose in Storyline	Lesson 1 gave the class the five-material test (copper bright; glass, dry wood and plastic off; diode lit forward and off in reverse) and the first class model of electrons. This lesson turns those yes/no results into a quantitative classification using resistivity and links it to Mr Otieno's wiring. It also leaves open WHY the resistivities differ so much (energy bands, taken up later) and pushes the class toward the semiconductor question taken up in Lesson 3.
Safety Notes	Do not experiment with mains electricity (240 V) in class; all circuit work uses the 3 V supply (two 1.5 V dry cells). Keep dry cells away from heat and do not short-circuit them with crocodile clips or copper wire for more than a moment because wires and cells can get hot. Remind learners never to touch damaged or exposed household cable or sockets and never to use wet hands near electrical appliances. Report cracked insulation to an adult.

B. LESSON OVERVIEW

Ms Achieng opens by revisiting the Lesson 1 results: in the 3 V test circuit only copper lit the bulb, while glass, dry wood and plastic did not, and the diode conducted one way only. Learners predict how Mr Otieno's copper-and-PVC cable in a Nairobi house stays safe at 240 V, and where wood, glass and silicon would rank in a list from best to worst conductor. In the Observe phase, groups quickly re-run the fixed test (copper, glass, wood, plastic, diode) and then receive a resistivity data card (copper 1.7×10^{-8} , germanium 0.46, silicon 2.3×10^3 , dry wood about 10^8 to 10^{11} , glass about 10^{10} to 10^{14} , PVC about 10^{13} or more ohm metre). They order the materials, notice the huge gaps between powers of ten, calculate that silicon is about 1.4×10^{11} times more resistive than copper, and sort the materials into conductor, semiconductor and insulator. In the Explain phase they use the data to explain why household wiring uses copper covered in PVC, and why insulators allow almost no current rather than none. They add new questions to the DQB and revise the class model with a resistivity scale and a note that the reason for the differences is still unexplained.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the Lesson 1 results (copper lit the bulb; glass, dry wood and plastic did not; the diode conducted one way). Individually they write and sketch a prediction in their notebooks: (a) Rank copper, glass, dry wood, plastic and silicon from best to worst at letting current flow, and (b) Mr Otieno holds the PVC-covered copper cable carrying current in a Nairobi house at 240 V. Why is he safe? They then compare predictions with a partner and star where they disagree. Wanjiku predicts that wood is the 'second best' because 'it comes from living things', while Kamau predicts silicon will be near copper. Learners are invited to treat these as ideas to test.	Project the Lesson 1 results table and say: 'Last lesson our bulb told us yes or no. Today we ask: how much better is copper than glass, and where does silicon fit?' Give 3 minutes of silent writing, then say: 'Rank the five materials and explain Mr Otieno's safety using the word electrons.' WAIT TIME: 10-15 seconds of silence after posing the question before any talk. Pair-share for 2 minutes. Cold-call 4 pairs: 'Which material did you put last, and why?' Record ALL rankings on the board without judging. Ask: 'Who ranked differently from this pair? What was your reasoning?' Circulate and note learners who think insulators have exactly zero conduction or that thick wires matter more than material.	Eliciting prior ideas through a Predict-Explain-Compare routine. Making predictions public exposes ranking ideas and preconceptions (for example, wood is a better conductor than glass, or insulators conduct nothing at all) so that later data can confirm or challenge them, which builds learners' ownership of the revision.	Observe whether predictions include a ranking with a reason (not just a list). Look for: (1) copper placed first by at least 80% of learners; (2) use of electrons in explanations; (3) learners who connect the PVC cover to stopping current reaching the hand. Tally the misconceptions heard (for example, 'insulators have no electrons', 'silicon is the same as copper') on a sticky note to revisit in the Explain phase.
Observe Phase  VIDEO: 02 Conductors Insulators And Semi Conductors Source: Seavuria  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice Source: CK-12  Search ARES for similar readings	Part A (10 minutes): In groups of four, learners re-run the Lesson 1 test circuit (3 V from two 1.5 V dry cells, 3 V torch bulb, crocodile clips) in the fixed order: copper wire 10 cm (bulb bright), glass piece (off), dry wood ruler or stick (off), plastic strip from stripped cable insulation (off), silicon diode 1N4007 forward-connected (lit) and reverse-connected (off). They record bulb status in a table. Part B (20 minutes): Each group receives a resistivity data card with values at about 20 C in ohm metre: copper 1.7×10^{-8} ; germanium 0.46; silicon (pure) 2.3×10^{-3} ; dry wood about 10^8 to 10^{11} ; glass about 10^{10} to 10^{14} ; PVC about 10^{13} or more. They (1) cut and arrange cards on	Brief safety reminder: 'We only use our 3 V cells. We never test with the mains at home or in school.' Ask groups to assign roles: tester, recorder, reporter, safety checker. Say: 'Predict first: will the bulb light? Thumbs up or down. Then test.' Circulate and ask: 'Which material did the bulb tell you is a conductor? Does the bulb tell you how good a conductor?' Hand out the data cards after Part A and say: 'The bulb only says yes or no. Scientists measure resistivity to say how much.' Point to the number line: 'Look at the gap between copper and germanium, and between silicon and wood. What do you notice?' WAIT TIME: 10-15 seconds after the question. Cold-call 3 groups to state their	Data patterns and scale reasoning: ordering values across many powers of ten and calculating the ratio of silicon to copper gives learners a concrete sense of how huge the differences are. Sorting into three columns is a classification activity in which learners build categories from evidence (the numbers themselves) instead of memorising a definition, so the classification becomes a conclusion supported by the data.	Check the recorded table matches the fixed results: copper bright; glass, wood and plastic off; diode forward lit and reverse off. Check the number line order is copper, germanium, silicon, wood, glass and PVC. Check that learners calculate about 1.4×10^{11} and state it in words. Look for learners sorting copper under conductor, germanium and silicon under semiconductor, and wood, glass and PVC under insulator. Listen for the statement 'almost no current' rather than 'no current'.

	a number line of powers of ten from lowest to highest ρ, (2) calculate conductivity $1/\rho$ for copper (about 6×10^7 per ohm metre) and for silicon (about 4×10^{-4}), and (3) calculate how many times silicon is more resistive than copper (2.3×10^3 divided by 1.7×10^{-8}, about 1.4×10^{11}). They then sort materials into three columns: conductor, semiconductor, insulator. They compare their order with their Predict ranking and mark where it changed.	order. Ask: 'Why is the bulb dark for dry wood although wood is not infinite in resistivity?' (Guide: the current is too small to light the bulb, but not exactly zero.) Challenge: 'Each step on your number line is a factor of ten. How many steps from copper to PVC?' (More than 20.)		
Explain Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work in their groups to build a claim-evidence-reasoning (CER) explanation for the phenomenon: 'Why does Mr Otieno use copper cable covered in PVC for the sockets in a Nairobi house?' Claim: copper is a conductor and PVC is an insulator. Evidence: resistivity values (copper 1.7×10^{-8} ohm metre; PVC about 10^{13} or more ohm metre), the bulb results from the test circuit, and the 1.4×10^{11} comparison for silicon. Reasoning: current flows easily through copper (low ρ, less energy wasted as heat in the 2.5 mm squared cable), while PVC allows almost no current so the current stays in the wire, even at 240 V, and a person holding the insulated cable does not get a dangerous current. Learners also explain what silicon being in between means and why it is neither a good wire nor a good cover. They discuss the safety case: damaged PVC insulation, wet hands (water with salts reduces the protection), and why Kenyan homes use copper rather than dry wood or glass for wires. Each group writes one open	Say: 'Use the sentence frame: I claim ____ because the data show ____, which means ____.' Give 5 minutes of group writing. Cold-call 4 groups, one at a time: 'Read your claim and evidence.' After each, ask the others: 'What evidence would you add or challenge?' WAIT TIME: 10-15 seconds before each reply. Press for precision: 'You said insulators do not conduct. What does the resistivity of PVC (about 10^{13}) tell us about that word? Say it with almost.' Probe: 'If copper is such a good conductor, why not wear it as a cover for the wire?' (Guide: the current would pass through the person.) Ask: 'Why is a bulb dark with wood even though wood is not at the very top of the list?' Return to the Predict misconceptions: 'Who changed their ranking? What data changed your mind?' Also ask: 'Our data tell us WHAT is different. Do they tell us WHY? What do we still need to explain?' Record the open questions to move to the DQB. Do not give the energy band explanation yet.	Claim-Evidence-Reasoning (CER) with whole-class critique. Structuring the explanation forces learners to connect numerical evidence to the safety phenomenon, and peer challenge strengthens reasoning. Naming what the data cannot yet explain (why resistivities differ) creates a purposeful gap that motivates later lessons on energy bands and semiconductors.	Use a 3-point rubric on the CER: (1) claim uses the terms conductor and insulator correctly; (2) evidence quotes at least two resistivity values with units (ohm metre); (3) reasoning links low ρ to efficient current flow in copper and high ρ to safety in PVC and uses almost no current. Look for a learner who explains that silicon is a semiconductor with resistivity in between and cannot be used as either wire or cover. Collect one exit ticket sentence: 'PVC keeps Mr Otieno safe because ____.'

	question they cannot yet answer (for example, 'Why is copper's resistivity so low, and why do glass and PVC have such high resistivity when all are made of atoms?').			
Driving Question Board (DQB) Creation  VIDEO: 02 Conductors Insulators And Semi Conductors Source: Seavuria  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice Source: CK-12  Search ARES for similar readings	Learners go to the class DQB and (1) review the Lesson 1 questions, placing a green dot next to any question the resistivity data helped to answer (for example, 'Which materials let current flow?' and 'How can Mr Otieno hold the wire safely?'), (2) write new questions on sticky notes (for example, 'Why is copper's resistivity so much lower than glass?', 'Why does silicon conduct a little, and does temperature change it?', 'Why does the diode conduct only one way?'), and (3) cluster them under the headings Conductors and Insulators, Semiconductors, and Applications. Wanjiku and Kamau add the question: 'If all materials have electrons, why do some not let them move?' Each learner reads one question to a partner and says which later lesson might answer it.	Say: 'Look at our DQB. Which questions can we now answer with evidence from today? Place a green dot and say the evidence.' Give 4 minutes for writing new questions and say: 'Write questions about WHY, not just WHAT.' WAIT TIME: 10-15 seconds while learners think before cold-calling 3 learners to read a new question aloud. Ask: 'Which cluster does this belong to? Is it a question about the data, the cause or an application?' Group similar questions and model rewriting a vague one: 'Why is silicon special?' becomes 'Why does silicon conduct a little and more when warmer?' Link to the driving question: 'Which of these questions will help us explain the safety and efficiency of wiring, solar panels, computers and MRI machines?' Do not answer the why questions; mark them as coming next.	Question clustering on a public Driving Question Board (DQB). Tracking questions and evidence lets learners see their own progress and shows how new investigations will address unresolved questions, keeping the whole unit connected to the driving question.	Check that each learner has at least one new question and that at least half are 'why' or 'how' questions. Check green dots are accompanied by a stated piece of evidence (for example, 'PVC about 10^{13} ohm metre'). Note whether questions mention electrons, energy or temperature, which shows readiness for the later lessons. Record which learners still ask only 'what' questions for follow-up support.
Model Building Phase  VIDEO: 07 Conductors And Insulators <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Circuit Construction Kit:	Learners revise the class model, the cumulative drawing of electrons, atoms and energy bands begun in Lesson 1. In groups, they add: (1) a resistivity scale (a vertical or horizontal line with copper 10^{-8} , germanium 0.46, silicon 10^3 , wood and glass 10^8 to 10^{14} , and PVC at 10^{13} or more), (2) three labelled zones, conductor, semiconductor and insulator, with an example of each from the test circuit, (3) a sketch of Mr Otieno's cable showing a copper core with arrows for current	Say: 'Our model must now show how good or poor each material is, not just yes or no.' Provide chart paper and markers. Ask: 'What did we add from today's data? Show where the evidence is in your drawing.' Circulate with prompts: 'Where on your scale does silicon go? What does that say about how well it conducts?' and 'How does your drawing show why the PVC is safe?' WAIT TIME: 10-15 seconds before asking a group to justify a change. Cold-call 3 groups to explain one revision and one	Cumulative model revision with gallery critique. Adding evidence to the class model, and marking explicitly what it cannot yet explain, helps learners see models as tools that improve with data. Peer 'I like, I wonder' feedback encourages evidence-based critique.	Look for models that include: a scale with correct order (copper, germanium, silicon, wood, glass, PVC); the three categories correctly assigned; a labelled cable with copper and PVC; and an honest question mark for the unexplained cause. Listen for learners using almost no current. Use a quick self-check: learners show 1-4 fingers for how confident they are in explaining why wiring uses copper with PVC. Follow up with those showing 1-2.

DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	flow and a PVC cover with a label 'almost no current escapes', and (4) a '?' sign beside the idea 'why do electrons move easily in copper but hardly at all in PVC?' to show an unfinished part of the model. Each group presents its revised model to another group, which gives one 'I like' and one 'I wonder' comment. Finally, learners complete the first two columns of the summary table (What we observed, What we learned) and start the third (What we can explain).	remaining question. Praise any group who marks a '?' for the cause of the differences. Close: 'We can say WHAT is different: resistivity differs by more than 20 powers of ten. Next we ask WHY, starting with silicon and germanium.' Collect the summary table drafts.		
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D. TEACHER REFLECTION

1) Did learners treat the resistivity numbers as evidence, or did they only copy them down? Which questions helped them notice the very large gaps between powers of ten? 2) How many learners still say insulators have zero conduction or use no current? Plan to revisit with 'almost no current' in the next lesson's opening. 3) Did learners separate WHAT (classification) from WHY (the energy gap explanation)? Did the open questions on the DQB point to the energy bands and semiconductors? 4) Which groups struggled with powers of ten or with the $1/\rho$ calculation and need support? 5) Were all safety rules respected (3 V only, no short circuits, no discussion of experimenting with mains)? 6) How did Kenyan contexts such as Mr Otieno's wiring in Nairobi help learners connect to the phenomenon? What local examples (for example, a jua kali electrician or a damaged extension cable) could be added?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We re-ran the test circuit (3 V, bulb): copper bright; glass, dry wood and plastic off; silicon diode lit forward and off in reverse. The resistivity card showed copper about 1.7×10^{-8} ohm metre, germanium 0.46, silicon 2.3×10^3 , dry wood about 10^8 to 10^{11} , glass about 10^{10} to 10^{14} and PVC about 10^{13} or more. Silicon is about 1.4×10^{11} times more resistive than copper.
What did I learn?	Materials can be classified by resistivity: conductors (copper) have very low resistivity; semiconductors (germanium and silicon) are in between; insulators (wood, glass, PVC) have very high resistivity and allow almost no current. Conductivity is $1/\rho$. Household wiring uses copper for efficient current flow and PVC to keep current inside the wire so people can handle cables safely.
How does this explain the phenomenon?	We can explain why Mr Otieno uses copper cable covered in PVC for a Nairobi house at 240 V: copper's low resistivity carries current efficiently and PVC's very high resistivity almost stops current reaching a person. We cannot yet explain WHY the resistivities differ so much or why silicon conducts a little, which we will investigate next.

LESSON 3 (40 minutes): Intrinsic Semiconductors: Silicon and Germanium

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students investigate why pure (intrinsic) silicon and germanium conduct a little, and why their conductivity rises when they are warmer, using energy gaps and resistivity data to revise the class model.
Knowledge	Intrinsic means pure. Silicon (Eg 1.1 eV, rho about 2.3×10^3 ohm metre) and germanium (Eg 0.67 eV, rho about 0.46 ohm metre) have a small energy gap, so at room temperature a few electrons gain enough energy to become free electrons. Each freed electron leaves a hole, making an electron-hole pair. Higher temperature frees more pairs, so conductivity rises (resistivity falls). Copper, a metal, behaves differently: its resistance rises slightly with temperature. Germanium has the smaller gap and so is less resistive than silicon.
Skills	Developing and using models (NGSS SEP 2): revise the class model of electrons, atoms and energy bands to show electron-hole pairs. Analysing and interpreting data (SEP 4): compare gaps and resistivities and read a temperature trend. Constructing explanations (SEP 6): explain with evidence why silicon conducts a little and more when warm. Communication and collaboration: share ideas in pairs and with the class.
Attitudes	Curiosity about why the same atoms behave differently. Responsibility when handling warm water and electrical equipment. Respect for peers' ideas when revising models. Appreciation of how semiconductor science supports Kenyan technology such as phones and solar kits.
Key Inquiry Question	How do semiconductors and superconductors differ from ordinary conductors and insulators? (This lesson addresses pure semiconductors.)
Purpose in Storyline	In Lesson 1 students saw the bulb light for copper only and the diode conduct one way, and made a first class model. Lesson 2 classified materials by resistivity into conductors, insulators and semiconductors. Lesson 3 explains why silicon and germanium sit in between, by adding an energy gap and electron-hole pairs to the class model, and shows that temperature changes conduction. This prepares Lesson 4, where students ask how doping gives control, and answers part of the DQB question about what makes the semiconductor in between so useful.
Safety Notes	Use only the 3 V supply (two 1.5 V dry cells). Never use mains electricity in class. Warm water for the temperature activity must be hand-hot (no hotter than about 50 C), poured by the teacher; keep the diode and clips dry, using a sealed plastic bag if needed. Do not touch hot water containers without a cloth. Dry hands before touching circuits. Dispose of nothing electrical in water. Wash hands after the lesson.

B. LESSON OVERVIEW

Ms Achieng reminds the class of Mr Otieno's copper cable in PVC in the Nairobi house, and of Lesson 2's resistivity order: copper lowest, then germanium, then silicon, then wood, then glass and PVC. Wanjiku and Kamau ask why silicon is not an insulator, since its atoms hold electrons just like glass. Students predict how a silicon semiconductor will behave when it is warmed, then observe evidence: an energy gap and resistivity data card for silicon, germanium, copper and glass, plus a demonstration using the 3 V test circuit with the 1N4007 silicon diode, forward-connected, where the bulb lit in Lesson 1, and a class analysis of a temperature trend dataset for pure silicon. Students then explain, using the energy gap idea, why a few electrons are freed and how holes appear. They add questions to the Driving Question Board (DQB) and revise the class model by adding a small gap, electron-hole pairs and the effect of temperature.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: electronics intrinsic semiconductor silicon  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: electronics intrinsic semiconductor silicon	Students look at the Lesson 2 resistivity order on the board (copper 1.7×10^{-8} , germanium 0.46, silicon 2.3×10^{-3} , dry wood 10^8 to 10^{11} , glass 10^{10} to 10^{14} , PVC 10^{13} or more ohm metre). Ms Achieng poses: 'Pure silicon is much more resistive than copper but far less resistive than glass. In Nairobi at midday the room is warm; at night it is cool. Predict what happens to the conduction of a pure silicon piece when it is warmed.' Each student silently writes a prediction in their notebook with a quick sketch of silicon atoms and electrons, and finishes the sentence: 'I think warming silicon will make it conduct ___ because ____.' Pairs then compare, and note whether they think copper will behave the same way.	Display the resistivity order. Say: 'Remember Mr Otieno's copper cable and PVC cover. Silicon sits in between. Before I give you any new information, I want your best thinking.' Give 2 minutes of silent writing, then 1 minute pair talk. WAIT TIME: after asking 'What do you think happens when silicon is warmed?', wait 10-15 seconds before taking any answer. Cold-call 4 pairs (not volunteers) and record predictions as tallies on the board: more, less, same. Ask 'What are you assuming about the electrons in silicon?' Do not say which is right. Say: 'Keep your prediction; we will test it with evidence.'	Predict-Observe-Explain with public commitment: writing a prediction and sketching the electron picture before instruction surfaces ideas (for example that all electrons in silicon are stuck or all are free) so students can later compare them with evidence, which makes the conceptual change visible and memorable.	Check notebooks for: (a) a clear prediction (more, less or same); (b) a reason that mentions electrons or energy; (c) whether students link silicon to Lesson 2's in-between resistivity. Look for the misconception that warming makes silicon behave like a metal (resistance rises) or that insulators and semiconductors are the same. Record the tally on the board and note which students gave no reasoning, for follow-up in the Explain phase.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: electronics intrinsic semiconductor silicon  READING: <i>No closely matching reading in the ARES library for this activity.</i>	Students gather evidence in three parts (about 14 minutes). Part A: Each group receives an evidence card with energy gaps at about 300 K (copper 0 eV, bands overlap; germanium 0.67 eV; silicon 1.1 eV; glass above about 5 eV) and resistivity values from Lesson 2. They rank materials by gap and by resistivity and record that the order matches: smaller gap, lower resistivity. Part B: Ms Achieng runs the Lesson 1 test circuit (3 V from two 1.5 V dry cells, 3 V torch bulb, crocodile clips) with the 1N4007 silicon diode forward-connected: the bulb lit. Then she warms the diode in a sealed bag in hand-hot water while	Distribute the evidence cards and circuit materials. Say: 'Wanjiku and Kamau are asking: what do the numbers tell us about why silicon is in between?' Circulate and ask: 'Which has the smaller gap, silicon or germanium? Which has the lower resistivity? What pattern do you see?' For Part B, demonstrate with the supply at 3 V only and remind: 'Warm water only, hand-hot, dry clips.' State the safe circuit rules. Ask: 'What do you notice about the bulb?' WAIT TIME 10-15 seconds before taking answers. Be honest that a torch bulb is a crude detector and a small change may be hard to see; say: 'A bulb is not sensitive, so we use the	Data analysis with pattern finding: ranking gap and resistivity side by side lets students notice the relationship themselves before the idea is named, and contrasting silicon and copper graphs builds the observation that temperature has opposite effects on semiconductors and metals.	Observe whether groups: (1) rank materials correctly (copper, germanium, silicon, glass) by both gap and resistivity; (2) state that silicon's conductivity rises with temperature using data (for example 'resistivity went down as temperature went up'); (3) correctly say copper's resistance rises slightly. Listen for confusion between resistivity and conductivity and prompt: 'Conductivity is the reverse of resistivity ($1/\rho$).' Note any group that reports the bulb as a precise measurement.

Search ARES for readings Search terms: electronics intrinsic semiconductor silicon	students watch the bulb and observe a small change. Students record what they see. Part C: Groups analyse a temperature dataset for pure silicon, showing resistivity falling as temperature rises, and a copper dataset showing resistance rising slightly with temperature. They sketch two graphs of conductivity against temperature (one for silicon rising, one for copper slightly falling).	dataset to see the trend clearly.' Cold-call 3 groups to describe the silicon trend and 2 groups to describe the copper trend. Do not yet explain why; say: 'Hold your explanation until the next phase.'		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: electronics intrinsic semiconductor silicon  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: electronics intrinsic semiconductor silicon	Students use the energy band picture to explain their observations. Ms Achieng draws two bands: a full valence band and an empty conduction band separated by a gap. Pairs explain: (1) why copper, with no gap (0 eV), conducts easily, (2) why glass, with a gap above about 5 eV, has almost no free electrons, and (3) why silicon (1.1 eV) and germanium (0.67 eV) have a few. Students learn that at room temperature a few electrons gain enough thermal energy to jump the gap and become free electrons, and each leaves behind a hole, forming an electron-hole pair; both can carry current. Warming gives more electrons enough energy, so more pairs form and conductivity rises. Because germanium's gap is smaller, more electrons cross at the same temperature, so germanium has lower resistivity than silicon. Students write a 3-sentence explanation using the words energy gap, free electron, hole, temperature and compare with their earlier prediction. They also explain that copper's resistance rises slightly with temperature but this is a different effect from the semiconductor trend, and note that pure silicon is	Draw the band diagram and say: 'Here is the gap. Where is silicon on this picture? Where is copper? Where is glass?' Use a think-pair-share: 1 minute of pair talk, then cold-call 4 pairs to explain one part each. WAIT TIME 10-15 seconds before each cold call. Probe with: 'Why does a free electron leave a hole behind?', 'Is the silicon charged after electrons are freed?' (answer: no, still neutral overall), and 'Why would germanium conduct more than silicon at the same temperature?' Use the phrase 'almost no' when discussing glass: say 'Glass conducts almost none, not zero.' Do not call semiconductors half conductors. Do not teach doping; if a student asks how to make silicon conduct better, say: 'Great question, put it on the DQB; Lesson 4 will investigate it.' Ask students to compare to their prediction and use the sentence frame: 'My prediction was ____ and the evidence shows ____ because ____.'	Claim-Evidence-Reasoning with a band diagram: students make a claim about temperature and conduction, support it with data from the evidence card and dataset, and reason with the energy gap model, which connects the observations to a particle-level mechanism and links to the Nairobi wiring phenomenon.	Check written explanations for: (a) a correct statement that a small gap lets a few electrons become free; (b) mention of both free electrons and holes as pairs; (c) correct link between higher temperature and more pairs and higher conductivity; (d) correct comparison of germanium and silicon by gap. Look for errors such as 'holes are positive particles that are added' or 'warming adds electrons to silicon'. Use a quick whiteboard check: 'Which gap is smaller, silicon or germanium, and which conducts more?' Expect at least most students to answer correctly; reteach with the band diagram for those who do not.

	still electrically neutral.			
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: electronics intrinsic semiconductor silicon  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: electronics intrinsic semiconductor silicon	Students revisit the Class DQB. Using sticky notes, each pair writes one question that this lesson answered and one new question it raised. Possible examples: 'How can we make silicon conduct much more?', 'Why does the diode conduct only one way?', 'Why do solar panels use silicon?', 'Do superconductors have a gap?', 'Does warming help or harm computers?' Pairs place notes under headings 'Answered so far', 'Partly answered' and 'Still to investigate'. They link the answered note about why silicon conducts a little to the original puzzle: why metals like copper carry current easily while plastic and wood do not, and what makes the semiconductor in between so useful.	Say: 'Wanjiku and Kamau keep the DQB. What can we now answer, and what do we still need to find out?' Give 3 minutes for writing. WAIT TIME 10-15 seconds after asking 'Which question is still puzzling you most?' Cold-call 3 pairs to read and place a note. Group similar questions and point out the doping and diode questions: 'These will be the focus of Lesson 4.' Move the question 'Why do some materials conduct a little?' to 'Partly answered' and ask students to justify it with evidence from today. Make sure at least one note asks about practical uses such as phones, M-Pesa or solar kits.	Question tracking on a shared board: sorting questions by how far they are answered makes progress visible, links each lesson to the driving question and motivates the next investigation by turning curiosity into a coherent path.	Check that notes are specific (for example 'Why does doping increase conductivity?' rather than 'What is a semiconductor?'), that pairs sort at least one note correctly, and that new questions build on today's idea of electron-hole pairs and temperature. Count how many pairs wrote a question linked to a Kenyan application. Note students who only restate facts rather than ask a question.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: electronics intrinsic semiconductor silicon  READING: No closely matching reading in the ARES	Students revise the class model of electrons, atoms and energy bands made in Lesson 1. Using coloured markers, each group adds to the class model: (1) a small energy gap for silicon (1.1 eV) and germanium (0.67 eV) compared with copper (0 eV) and glass (above about 5 eV); (2) a few electrons jumping the gap to become free electrons at room temperature; (3) a hole left behind in the valence band, shown as an electron-hole pair; (4) more pairs when temperature increases, with an arrow labelled 'warmer, more pairs, higher conductivity'; and (5) a note that copper's resistance	Display the class model and say: 'Our model started in Lesson 1 with just electrons and atoms. What must we add today so that it explains silicon?' Give 8 minutes for groups to revise. Circulate with: 'Where in your model is the hole? What makes more pairs?' WAIT TIME 10-15 seconds after asking 'What is still missing from our model?' Cold-call 3 groups to explain one change each, and invite another group to challenge with evidence. Confirm that the model correctly shows neutral silicon and the same fixed numbers (1.1 eV, 0.67 eV). End by saying: 'Pure silicon conducts a	Cumulative model revision: adding new elements to a shared model and stating limitations shows students how scientific explanations improve as evidence accumulates, and gives a visible representation they can reuse to explain the phenomenon in later lessons.	Look for models that show: a gap smaller than the insulator gap and larger than zero; electrons and holes created together; more pairs at higher temperature; copper treated differently. Check captions use the conventions (conductor, insulator, semiconductor, intrinsic, free electrons, holes). Use an exit prompt: 'In one sentence, why does warming increase the conduction of pure silicon?' Success looks like 'warming gives more electrons enough energy to cross the gap, forming more electron-hole pairs.'

<i>library for this activity.</i> Search ARES for readings Search terms: electronics intrinsic semiconductor silicon	rises slightly with temperature. Groups write a short caption linking the model to the Nairobi house: 'Copper carries current and PVC insulates; silicon is in between and controlled by temperature.' They also record a model limitation: pure silicon conducts too little to be very useful, so what could change that?	little and more when warmer. In Lesson 4 we ask how we can control it.' Collect the group captions.		
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D. TEACHER REFLECTION

After the lesson, consider: Did students use the energy gap rather than just stating that warming makes silicon conduct more? Which students held the idea that holes are real particles added to silicon or that silicon becomes charged, and how did the band diagram change their thinking? Was the bulb change too small to see, and did students correctly rely on the dataset instead? Did students keep intrinsic silicon distinct from doped silicon, leaving doping for Lesson 4? Which DQB questions show readiness for doping and diodes? How well did cold-calling and 10-15 seconds of wait time bring in quieter students? What would you change in the safety routine with warm water?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the silicon diode (1N4007) forward-connected lit the 3 V bulb, as in Lesson 1, and that the data show silicon (rho about 2.3×10^3 ohm metre) and germanium (about 0.46 ohm metre) are far more resistive than copper (1.7×10^{-8} ohm metre) but far less than glass. The temperature dataset showed the resistivity of pure silicon falling as it was warmed, while copper's resistance rose slightly.
What did I learn?	We learned that pure (intrinsic) silicon and germanium have a small energy gap (1.1 eV and 0.67 eV), so at room temperature a few electrons gain enough energy to become free electrons, each leaving a hole to make an electron-hole pair. Germanium, with the smaller gap, conducts more than silicon. Warming creates more pairs so conductivity rises.
How does this explain the phenomenon?	We explained that silicon conducts a little because its small gap lets a few electrons cross, unlike glass, wood and PVC with a gap above about 5 eV that allow almost no free electrons, and unlike copper, where the bands overlap. This helps explain why semiconductors sit between copper wire and its plastic cover in Mr Otieno's Nairobi house, and raises the question of how to control their conduction, which we will investigate next.

LESSON 4 (40 minutes (double period of 80 minutes recommended if the model-building phase is extended)): Doping: Making n-type and p-type Semiconductors

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students explain how doping pure silicon with phosphorus or boron makes extrinsic n-type or p-type semiconductors, and why this control makes semiconductors useful, including how a p-n junction gives the diode its one-way action in the bulb test.
Knowledge	Doping adds a very small number of impurity atoms (about 1 atom in 10^7 to 10^8) to pure (intrinsic) silicon. Phosphorus (group 5, 5 valence electrons) gives one extra free electron per atom, making n-type silicon with electrons as majority carriers. Boron (group 3, 3 valence electrons) leaves one missing electron, a hole, per atom, making p-type silicon with holes as majority carriers. Doped silicon is still electrically neutral overall. Doping raises conductivity by a factor of many thousands in a controlled way. Joining p-type and n-type gives a p-n junction, a diode that conducts when forward-biased (bulb lit) and not when reverse-biased (bulb off).
Skills	Developing and using models (drawing atom-level pictures of n-type and p-type silicon); analysing and interpreting data (comparing conductivity of pure and doped silicon); constructing explanations (why the diode conducted one way only); communication and collaboration in pairs and groups.
Attitudes	Curiosity about how tiny changes in a material produce big changes in behaviour; responsibility when handling circuit components and sharing roles; respect for others' ideas when peer-reviewing models; appreciation of how electronics shapes daily life in Kenya (phones, M-Pesa, solar kits).
Key Inquiry Question	How do semiconductors and superconductors differ from ordinary conductors and insulators? How is electronics applied in day-to-day life?
Purpose in Storyline	Lesson 3 showed that pure silicon and germanium have a small energy gap (1.1 eV and 0.67 eV), so only a few electron and hole pairs form and conductivity rises with temperature. Students now ask: if pure silicon conducts so poorly (resistivity 2.3×10^3 ohm metre against 1.7×10^{-8} for copper), how can it be useful? Doping gives the answer and explains the diode in the Lesson 1 test: why the bulb lit in one direction only. This prepares Lesson 5 (testing the band model with a simulation) and Lesson 7 (solar panels and microchips).
Safety Notes	Use only the low-voltage 3 V circuit (two 1.5 V dry cells) with the 1N4007 diode; never connect the diode or any component to mains 240 V. Check that clips do not short-circuit the cells (cells and wires may get warm). Do not touch exposed wire ends of damaged cable. Dispose of used cells at a collection point, not in the open. No chemicals are handled; doping is modelled with counters and drawings only.

B. LESSON OVERVIEW

Ms Achieng opens by returning to the diode result from the five-material test: the bulb lit when the 1N4007 was forward-connected and stayed off when reverse-connected. Students first predict how a tiny amount of another element might change pure silicon. They then re-run the diode test in both directions to recollect the evidence, and use a counter-based bead model (a grid of silicon atoms with phosphorus or boron substitutions) to see where extra free electrons and holes come from. In the Explain phase they use the doping ratio (1 atom in 10^7 to 10^8) and the claim that conductivity rises many thousands of times to argue why doping is useful, and they connect a p-n junction to the one-way action of the diode. Students add new questions to the Driving Question Board (DQB) and revise the class model to include n-type and p-type regions. Throughout, the Nairobi electrician Mr Otieno's copper cable and the silicon chips in phones and the Garissa solar plant anchor the ideas.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  HTML: Doping Source: PhET  Search ARES for similar readings	Students recall the Lesson 1 test: copper lit the bulb; glass, dry wood and plastic did not; the silicon diode lit the bulb forward-connected and not reverse-connected. On a prediction sheet they individually answer: 'Pure silicon has resistivity about 2.3×10^3 ohm metre, about 10^{11} times more resistive than copper. If we replace just 1 silicon atom in every 10 million to 100 million with phosphorus or boron, what will happen to its conductivity? Sketch what you think the atoms and electrons look like.' They commit to one of three options: (a) no change, (b) small change, (c) very large change, and give a reason. They share with a partner, then a few share with the class.	Hold up the 1N4007 diode and say: "Remember Wanjiku and Kamau's test: the bulb lit one way and not the other. Today we find out why. Silicon is the main material inside this diode." Write the three options on the board. Say: "Take 2 minutes to write your prediction and draw it. There are no wrong predictions today, only ideas to test." Circulate and note common ideas (for example 'adding anything makes it worse' or 'impurities do nothing'). Then ask: "Turn to your partner and explain your reason in 30 seconds." Cold-call 4 pairs, giving WAIT TIME of 10 to 15 seconds after each question: "Why did you choose that option?" Do not say which is correct. Record the vote count (a/b/c) on the board.	Predict-Observe-Explain (POE) with a public vote: committing to a prediction makes learners' existing ideas visible, so that later evidence creates a genuine need to revise them. Sketching atoms and electrons (developing models) links the prediction to the class model begun in Lesson 1.	Check prediction sheets for: (1) a clear choice of a, b or c; (2) a reason that mentions electrons or atoms rather than only 'it is silicon'; (3) a sketch that shows electrons, atoms or gaps. Tally how many learners chose (c) very large change. Note learners who say impurities must reduce conductivity, who will need the evidence in the next phase.
Observe Phase  VIDEO: 06 P N Junctions And Diodes Source: Seavuria  Search ARES for similar videos  HTML: Doping Source: PhET  Search ARES for similar readings	Part A (10 minutes): In groups of 4 using the standard circuit (3 V from two 1.5 V dry cells in series, 3 V torch bulb, crocodile clips), students re-test the 1N4007 diode forward-connected (bulb lit) and reverse-connected (bulb off), recording in a table with bulb brightness and the direction of the diode's marked band relative to the cell. Part B (10 minutes): Students use a model grid of 25 'silicon atoms' (grey counters, each with 4 outer electrons shown as dots) and then (i) replace 1 counter with a 'phosphorus' counter (5 dots) and note the extra electron; (ii) replace another grid with a 'boron' counter (3 dots) and note the missing electron (a hole shown as an empty circle). They also read a short data card: 'Pure silicon: 2.3×10^3 ohm metre. Doping about 1	Distribute materials and say: "Roles: circuit builder, recorder, timekeeper, materials manager. Swap roles for Part B." Remind: "Only the 3 V cells. No mains. If a clip or cell feels warm, disconnect and call me." For Part A ask: "Which direction lit the bulb? How do you know the diode is reversed?" For Part B model the first substitution at one group's table: "Silicon has 4 outer electrons and shares them with 4 neighbours. Phosphorus has 5. How many are left over after bonding?" WAIT TIME of 10 to 15 seconds. Then: "Boron has 3. How many bonds can it complete, and what is missing?" Cold-call 3 groups to describe one observation each. Prompt: "Is the doped silicon charged overall? Count the protons and electrons."	Evidence gathering with a physical model (counter-based analogy) and a structured data table: learners see from the model where the free electron or hole comes from, while the data card gives the scale (1 in 10^7 to 10^8) and effect (thousands of times more conductive). Linking the concrete circuit result to the atom-level model supports claim-evidence-reasoning later.	Observable indicators: groups correctly record forward (lit) and reverse (off); counters show 1 extra dot for phosphorus and 1 empty circle (hole) for boron; learners say 'free electron' for n-type and 'hole' for p-type (conventions); they state that doped silicon stays neutral. Use a checklist to note groups that call a hole a 'positive particle added' rather than a missing electron, and address it in the Explain phase.

	atom in 10^7 to 10^8 with phosphorus or boron raises conductivity by a factor of many thousands.' They record observations: what moved, what stayed, and where the electron or hole is.	Listen for the idea that each added atom is neutral, so the doped silicon is still electrically neutral overall. Circulate with an observation checklist.		
Explain Phase  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  HTML: Doping Source: PhET  Search ARES for similar readings	Students work in their groups to write a claim-evidence-reasoning (CER) explanation answering: 'Why is doped silicon more useful in electronics than pure silicon, and why did the diode light the bulb in only one direction?' They must use: phosphorus and n-type (extra free electrons, electrons as majority carriers); boron and p-type (holes as majority carriers); the doping ratio (1 in 10^7 to 10^8) and the effect (conductivity thousands of times higher, controlled by how much dopant is added); and the p-n junction (a p-type piece joined to an n-type piece) as the structure of the diode. They draw a labelled picture of a p-n junction and use arrows to show that the bulb lights forward-biased and not reverse-biased. They then compare their explanation with another group using a feedback checklist and revise one sentence. Finally they re-open their Predict sheet and write, 'My prediction was ... The evidence shows ... So I now think ...'.	Say: "Use the evidence from the bulb and the counters, not just what you remember. Your claim must answer the question in one sentence." Give sentence frames on the board: 'Doping with phosphorus gives ____ so the majority carriers are ____.' 'Doping with boron gives ____ so the majority carriers are ____.' Ask the class: "Pure silicon is about 10^{11} times more resistive than copper. Why can we not just use copper in a diode?" WAIT TIME of 10 to 15 seconds, then cold-call 3 learners. Address the misconception directly: "A hole is not a new particle added. It is a missing electron. And doped silicon is still neutral." Then say: "Why is controlling the amount of dopant powerful? Think about a phone chip in Nairobi with billions of transistors." Cold-call 3 more learners. For the junction, say: "When the diode is forward-biased the bulb lit, when reverse-biased it was off. What does the junction let through, and what does it hold back?" Accept answers that carriers can cross in one direction only; do not go beyond the level of this lesson (details of the depletion region are not needed). Connect back: "Copper cable carries Mr Otieno's current in both directions; a diode lets it pass one way only. That is what makes it useful in solar panels and chips."	Claim-Evidence-Reasoning (CER) with peer critique: CER pushes learners to tie the claim to specific evidence (bulb results, counters, data card) and to scientific reasoning (carriers and structure). Peer feedback and the prediction revision make the change in thinking explicit, which strengthens conceptual change.	Look for explanations that: (1) correctly name phosphorus as n-type with extra free electrons and boron as p-type with holes; (2) state the majority carriers; (3) cite 1 in 10^7 to 10^8 and a large rise in conductivity; (4) state that doped silicon is still neutral; (5) link the p-n junction to forward (lit) and reverse (off) results. Mark each group against these five points. Learners who describe a hole as a positive atom or say doping makes silicon a conductor like copper need a short follow-up conversation. Ask 3 learners to give an exit-ticket sentence on why the diode works one way.

Driving Question Board (DQB) Creation  VIDEO: 06 P N Junctions And Diodes Source: Seavuria  Search ARES for similar videos  HTML: Doping Source: PhET  Search ARES for similar readings	Students return to the Class DQB, which holds questions from earlier lessons. In pairs, they (1) put a tick on questions now partly or fully answered by today's evidence (for example 'Why does the diode work in one direction only?' and 'Why is silicon useful?'), writing a short evidence note on a sticky note; and (2) add at least one new question arising from this lesson, such as 'How does a p-n junction make electricity in a solar panel at Garissa?', 'How much dopant is needed?', 'Can we dope germanium too?' or 'What happens when a diode is heated?'. They sort new questions into categories: Doping, Diodes and junctions, Solar and microchips, Superconductors and MRI.	Say: "Go to the DQB with your partner. Find one question that today's work helps answer and one new question you now have." Allow 5 minutes. Then ask: "Which question did we answer, and what is your evidence?" WAIT TIME of 10 to 15 seconds. Cold-call 3 pairs. Say: "Which new question will Lesson 5 (the simulation) help with? Which one must wait for Lessons 6 and 7?" Cold-call 2 pairs. Point out the link: "Several of you asked about solar panels. Garissa solar plant is about 50 MW and uses silicon p-n junctions. We will return to it in Lesson 7." Move any question that was answered to the 'Answered' column with the evidence note attached.	Driving Question Board as a public record of growing understanding: connecting new questions to evidence maintains learner ownership of the unit and shows which ideas still need investigation, which motivates the following lessons.	Observe whether sticky notes give evidence (for example 'bulb lit forward only') rather than just 'we learned it'. Check that new questions are specific to doping or junctions and not repeats of old ones. Count how many pairs add a question that links to Kenyan applications (solar kits, M-Pesa phones, Garissa plant), which shows transfer.
Model Building Phase  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  HTML: Doping Source: PhET  Search ARES for similar readings	Students revise the cumulative class model (the drawing of electrons, atoms and energy bands). Each group adds to the Lesson 3 model: (1) a panel for n-type silicon showing a phosphorus atom, one extra free electron, and the label 'majority carriers: electrons'; (2) a panel for p-type silicon showing a boron atom, a hole, and 'majority carriers: holes'; (3) a p-n junction drawn as p-type joined to n-type, with the diode symbol, and an arrow with 'forward-biased: bulb lit' and 'reverse-biased: bulb off'. They label where the doping ratio applies (about 1 in 10^7 to 10^8) and note that both pieces remain neutral overall. Groups then do a gallery walk, leaving one star (something clear) and one wish (something to improve) on each model.	Say: "Take the Lesson 3 class model. We are not starting again; we are adding to it. What must we add to explain the diode and why doping is useful?" Give 8 minutes for revision. Circulate and ask: "Where is the free electron in your n-type drawing? Where is the hole in the p-type? Is your doped silicon charged?" WAIT TIME of 10 to 15 seconds for each. Then say: "Gallery walk: visit two other groups. Leave one star and one wish." Cold-call 3 groups to share what they changed from their previous model and why, referencing evidence. Pin the best consensus version as the Class model for the board. Close by asking: "Our model now explains the diode. What does it not yet explain? How does it connect to energy bands?" Take 2 responses and note that Lesson 5 will test the band model with a simulation.	Cumulative model revision with peer gallery walk: revising the same model each lesson shows how scientific ideas grow with evidence. Star-and-wish feedback encourages learners to evaluate models against evidence and to identify gaps that lead to the next investigation.	Check each model for: n-type with phosphorus, extra free electron and electrons as majority carriers; p-type with boron, hole and holes as majority carriers; a p-n junction labelled with forward (lit) and reverse (off); neutrality stated; correct conventions (intrinsic, extrinsic, forward-biased, reverse-biased). Record which groups can say what the model cannot yet explain. Use a 3-point rubric: complete, mostly complete, needs support.

D. TEACHER REFLECTION

After the lesson, ask yourself: Did learners distinguish a hole as a missing electron rather than a particle, and did they keep in mind that doped silicon is still neutral? Which prediction option (a, b or c) did most learners choose at the start, and how did it change by the end? Did every group link the bulb result (forward lit, reverse off) to a p-n junction? Were the cold-call and WAIT TIME routines used so that quieter learners such as those who rarely speak were heard? What new DQB questions suggest that learners need more time before the Lesson 5 simulation? Note any learners who need support with the idea of majority carriers and plan a short revisit using the counters.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We re-tested the 1N4007 diode: the bulb lit when it was forward-connected (forward-biased) and stayed off when it was reverse-connected (reverse-biased). Using counters, we saw that replacing one silicon atom with phosphorus (5 outer electrons) leaves an extra free electron, and replacing one with boron (3 outer electrons) leaves a hole. The data card showed that doping about 1 atom in 10^7 to 10^8 raises the conductivity of silicon by a factor of many thousands.
What did I learn?	Doping makes extrinsic semiconductors from pure (intrinsic) silicon. Phosphorus gives n-type (majority carriers: electrons) and boron gives p-type (majority carriers: holes). Doped silicon is still electrically neutral overall. Doping lets us control conductivity, and joining p-type and n-type makes a p-n junction, which is the diode.
How does this explain the phenomenon?	Pure silicon conducts very poorly (resistivity about 2.3×10^3 ohm metre), but a tiny amount of phosphorus or boron adds free electrons or holes, so conductivity rises thousands of times in a controlled way. A p-n junction lets current pass when forward-biased (bulb lit) and blocks it when reverse-biased (bulb off), which is why doped silicon is the basis of diodes, solar panels like those at Garissa, and microchips in phones and computers.

LESSON 5 (40 minutes (one lesson; double period of 80 minutes if only one computer or tablet is available per group of 4)): PhET Simulation: Testing the Band Model with Digital Evidence

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a simulation as digital evidence to test and refine the class band model by changing the energy gap, the doping type and the temperature, and then use the evidence to explain why copper lights the bulb while glass, wood and plastic do not and why silicon sits in between. This lesson teaches the skill of testing a model, not a new concept.
Knowledge	Learners use the simulation to check ideas from Lessons 2 to 4: metals have an energy gap of 0 eV (bands overlap); silicon has 1.1 eV; germanium has 0.67 eV; insulators such as glass have a gap above about 5 eV, so almost no electrons can move. Raising the temperature frees more electrons in a semiconductor. Doping with phosphorus gives n-type (extra free electrons) and doping with boron gives p-type (holes), and doped silicon stays electrically neutral overall.
Skills	Developing and using models (NGSS SEP 2): compare model predictions with simulation output and revise the model. Planning and carrying out investigations (SEP 3): change one variable at a time (gap, temperature or doping type). Analysing and interpreting data (SEP 4): record and compare simulation readings in a table. Engaging in argument from evidence (SEP 7): judge how far a simulation can be trusted compared with the real bulb test. Digital literacy: use a simulation on a computer or tablet, working offline from the pre-loaded copy.
Attitudes	Curiosity and open-mindedness: willingness to revise a model when evidence disagrees. Responsibility: each group member takes a role (operator, recorder, reporter, checker). Respect and collaboration: listening to group members' predictions and giving reasons. Honesty: reporting results exactly as observed, even when they differ from the prediction.
Key Inquiry Question	1. How do semiconductors and superconductors differ from ordinary conductors and insulators? 2. How is electronics applied in day-to-day life? (This lesson focuses on question 1 for conductors, insulators and semiconductors, and uses digital evidence to test the band model that explains the Nairobi wiring phenomenon.)
Purpose in Storyline	Lessons 2 to 4 gave the class classification by resistivity, intrinsic conduction and doping. In Lesson 5 the class stops adding ideas and instead tests the band model, because real atoms and bands cannot be seen. The simulation lets learners change the gap, temperature and doping in seconds. The tested model then carries into Lesson 6 (superconductors and MRI magnets) and Lesson 7 (final model-based answer on solar panels, microchips and MRI).
Safety Notes	No hazardous materials. Seat learners so that cables from tablets or laptops are not across walkways. Keep drinks away from devices. Limit screen time to about 20 minutes and let learners look away from the screen between tasks. If devices are charged from the mains, only the teacher connects and disconnects chargers. If there are fewer devices, use groups of 4 to 5, or the teacher projects the simulation and learners take turns to operate it.

B. LESSON OVERVIEW

Ms Achieng reminds the class of the Nairobi house where Mr Otieno holds a PVC-covered copper cable safely while it carries current, and of the five-material bulb test (copper bright; glass, dry wood and plastic off; the silicon 1N4007 diode lit forward-biased and off reverse-biased). Learners first predict, in writing, what a band diagram will look like for a metal, silicon, germanium and an insulator, and what will happen when temperature rises or phosphorus or boron is added. Groups then use the PhET simulation as digital evidence: they change one variable at a time, record what they see in a table, and compare it with the class model and their predictions. In the Explain phase they decide which parts of the class model are supported and which need revision, and discuss the limits of a simulation compared with the real bulb test. They add new questions and partial answers to the Driving Question Board (DQB), and finally revise the class model on a poster so that it shows bands, gap sizes, free

electrons, holes and temperature effects.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 03 Intrinsic And Extrinsic Semi Conductors Source: Seavuria  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners look at the one-page class model from Lesson 4 and a blank set of four energy band boxes labelled copper, silicon, germanium and glass. Working alone for 3 minutes, each learner sketches what they expect: where the gap is, and how many electrons are in the upper band for each material at room temperature. They also write one-sentence predictions: (a) 'If I raise the temperature of silicon, the number of free electrons will ...' (b) 'If I add phosphorus, the extra electrons will be ...' (c) 'If I add boron, the holes will be ...' (d) 'If I make the gap bigger, such as for glass, the current will ...'. They then share predictions with a partner and star the one they are least sure about.	Hold up a 3 V torch bulb and the copper wire and say: "Remember Mr Otieno in Nairobi. He holds the PVC-covered copper cable while it carries current. Today we cannot see the inside of copper or glass, so we will test our band model with a simulation. First, what do YOU think it will show?" Ask learners to sketch silently. Give 3 minutes of silent writing. Say: "Do not look at your notes from Lesson 4 yet; I want your best thinking." Circulate and note common ideas. After partner talk, WAIT TIME: ask "Who can predict what happens to silicon when it gets warmer?" and wait 10 to 15 seconds before taking answers. Cold-call 4 learners from different groups and ask each "What made you think that?" Record 3 predictions on the board under the heading 'Predictions to test'. Do not say which is correct.	Predict-Observe-Explain (POE) with elicited prior ideas: writing predictions before using the simulation makes learners commit to ideas, so any mismatch with the digital evidence becomes a reason to revise the model rather than a surprise to ignore.	Check sketches for: (1) copper drawn with no gap or overlapping bands (0 eV); (2) silicon and germanium drawn with a small gap and silicon larger than germanium (1.1 eV and 0.67 eV); (3) glass drawn with a much larger gap (above about 5 eV). Tally how many learners predict that raising temperature increases free electrons in silicon. Note learners who say that insulators have 'no' conduction at all, and prompt them later to say 'almost none'. Note learners who think doped silicon becomes charged; this idea will be tested in the simulation.
Observe Phase  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice <i>Related topic: not</i>	In groups of 4 (operator, recorder, reporter, checker), learners open the pre-loaded PhET simulation on conductivity and energy bands (offline copy) and run three short investigations. Investigation 1 (gap): select or set the gap for a metal (0 eV), germanium (0.67 eV), silicon (1.1 eV) and an insulator (above about 5 eV), keeping temperature the same, and record in a table how many electrons appear in the upper band and whether the simulated bulb or	Before groups start, model the first step: "Watch me. I change ONE thing only, the gap. Everything else stays the same. Why do I do that?" Cold-call 2 learners. Distribute roles with cards. Say: "Operator, click. Recorder, write exactly what you see, not what you expected. Checker, say 'one variable' if the operator changes more than one. Reporter, be ready to share." Circulate and ask each group at least one probing question: "Where is the gap for	Controlled-variable investigation with a data table: changing one variable at a time (gap, temperature, doping) lets learners link each change to one effect, which makes the simulation function as evidence rather than animation. Comparing results with their predictions builds model-testing skill (SEP 2 and SEP 3).	Check tables for: metal 0 eV with high current; germanium 0.67 eV with more free electrons than silicon at the same temperature; silicon 1.1 eV with a few free electrons; insulator above about 5 eV with almost none; free electrons and holes in silicon increase as temperature rises; phosphorus gives extra free electrons (n-type, majority carriers electrons); boron gives holes (p-type, majority carriers holes). Indicators of skill: the group

<i>an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	current reading is high, low or almost none. Investigation 2 (temperature): choose silicon and raise the temperature in 3 steps, recording the number of free electrons and holes at each step; then choose a metal if the simulation allows and note what happens. Investigation 3 (doping): add phosphorus (n-type) and then boron (p-type) to silicon, and record the majority carriers (electrons or holes), the current reading and whether the material is still neutral overall. Each group fills in a 3-column table and draws one labelled band diagram for each doped case. The checker confirms that only one variable was changed at a time.	copper in your picture? What does that mean for Mr Otieno's cable?" and "What is different between silicon at the first and the third temperature step?" Use WAIT TIME of 10 to 15 seconds after each question before prompting. At the 10-minute mark call: "Stop. Look away from the screen. Stretch your neck for 20 seconds." For groups that finish early, ask them to find the gap size at which the simulated current becomes almost none and to record that value. If devices are limited, project the simulation and let each group send an operator to change the setting while the others predict aloud first.		changes one variable at a time (checker calls it out), recorders write observed values rather than expected ones, and reporters use the terms 'free electrons', 'holes', 'energy gap'. Note groups that mix up n-type and p-type and ask them to point to the phosphorus (5 valence electrons) and boron (3 valence electrons) in the diagram.
Explain Phase  VIDEO: 02 Conductors Insulators And Semi Conductors Source: Seavuria Search ARES for similar videos  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Groups compare their prediction sketch with the simulation evidence and mark each prediction as 'supported', 'not supported' or 'needs more detail'. They then complete the claim-evidence-reasoning (CER) frame: Claim: 'Copper lit the bulb in Mr Otieno's test circuit while glass, wood and plastic did not because ...'; Evidence: simulation readings for gap 0 eV versus above about 5 eV; Reasoning: how the size of the energy gap controls how many electrons can move. They extend the CER to the diode: 'The silicon diode conducts only when forward-biased because ...' using the doping evidence (n-type and p-type) and what they know from Lesson 4 about the p-n junction giving one-way action. Finally, in a whole-class discussion they list two things the simulation shows well and two limits (for example it does not show real atoms, the numbers are simplified, and it	Say: "Look at your prediction and your table. Mark each prediction: supported, not supported, or needs more detail." Give 3 minutes. Then ask: "Whose prediction was NOT supported? That is the most useful result today." Cold-call 3 groups. Model the CER sentence frame on the board and give 5 minutes for groups to write. Ask: "Mr Otieno holds the PVC cable safely. Use your evidence to explain why. Use the words energy gap and almost none." WAIT TIME of 10 to 15 seconds before taking responses. Cold-call 4 learners, asking another learner to restate each answer in their own words. Challenge misconceptions with: "If insulators have a huge gap, do they have zero conduction? What does your simulation say?" (Guide to 'almost none'.) Ask: "If we add phosphorus to silicon, is the silicon now negatively charged? Use evidence." (Guide to: still	Claim-Evidence-Reasoning (CER) combined with model evaluation: learners must connect each claim about the Nairobi wiring phenomenon to specific simulation data and to the band model, which builds argument from evidence (SEP 7). Discussing the limits of the simulation develops the habit of judging how far any model or digital evidence can be trusted.	Look for CER answers that: (1) claim that copper conducts and plastic does not; (2) cite evidence such as 0 eV versus above about 5 eV; (3) reason that when the gap is large almost no electrons can move so PVC is a safe cover. Strong responses state that silicon (1.1 eV) and germanium (0.67 eV) conduct a little, more when warmer, and that phosphorus (n-type) and boron (p-type) raise conductivity in a controlled way. Check learners avoid 'zero conduction' for insulators and 'charged' for doped silicon. Ask 3 learners to give a thumbs-up, sideways or down for 'I can use the simulation to test a model' and note any thumbs-down for follow-up in Lesson 6.

	cannot show the bulb or the 3 V dry cells). They decide which parts of the class model to keep, change or add.	electrically neutral overall, since each phosphorus atom is also neutral.) Then ask: "A simulation is not the real world. Name one thing it did not show that the bulb test did." Take 3 answers. Record the agreed limits on the board. Do NOT re-teach doping or intrinsic conduction; only point back to the earlier lessons if a group is stuck.		
Driving Question Board (DQB) Creation  VIDEO: 01 Conductors Insulators And Semi Conductors Source: Seavuria  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revisit the class DQB with the driving question: 'Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines?' In groups they move sticky notes: they place questions that the simulation helped to answer (for example, 'Why does silicon conduct more when warm?' and 'What is the difference between n-type and p-type?') under 'Partly answered', and write the evidence on a new note, such as 'PhET: silicon free electrons rose at three higher temperatures'. They also write at least one new question the simulation raised, such as 'How does a p-n junction make a solar panel produce electricity?', 'What happens to resistance at very low temperatures?' or 'How can an MRI magnet carry large current without losing energy?', and add it to the DQB.	Point to the DQB and say: "Let us check our board. Which of our questions did today's evidence help answer, even a little?" WAIT TIME of 10 to 15 seconds. Cold-call 3 learners to move a question and justify it with one piece of evidence from their table. Say: "A good note names the evidence, for example 'In the simulation, silicon at the highest temperature had more free electrons.'" Ask each group to write one NEW question and read it aloud; say: "Which of these questions will Lesson 6 and Lesson 7 help us answer?" Flag the questions about low temperature and MRI magnets for Lesson 6, and solar panels, microchips and the Garissa solar plant (50 MW) for Lesson 7. Do not answer them yet; say: "Keep them on the board. We will come back with evidence."	Driving Question Board as a living record: moving questions between 'unanswered', 'partly answered' and 'answered', and attaching evidence to each move, makes the growth in understanding visible and gives learners ownership of the next steps in the investigation.	Check that notes moved to 'Partly answered' name specific simulation evidence (not just 'because the teacher said so'). Count the new questions: aim for at least one per group, with at least one linked to superconductors or MRI (Lesson 6) and one to solar panels or microchips (Lesson 7). Note any group whose new question repeats an earlier one and prompt them to sharpen it with 'How' or 'Why' and a Kenyan context, for example Kenyatta National Hospital or the Garissa solar plant.
Model Building Phase  VIDEO: 03 Intrinsic And Extrinsic Semi Conductors Source: Seavuria	Learners return to the cumulative class model (the drawing of electrons, atoms and energy bands) started in Lesson 1 and revised in Lessons 2 to 4. In their groups they add and revise: (1) four band diagrams side by side for copper (0 eV, bands overlap),	Say: "Look at our class model. Does it explain why Mr Otieno can safely hold the PVC-covered copper cable? What must we ADD or CHANGE after today's evidence?" Give 8 minutes for group drawing. Say: "Use a colour pen for every change you make	Cumulative model revision with evidence-tagged changes: colour-coding each revision to a piece of evidence trains learners to justify changes to a model, while the gallery walk provides peer critique. This keeps the model tied to the Nairobi wiring phenomenon and	Check each group model for: copper with overlapping bands (0 eV); germanium (0.67 eV) with a smaller gap than silicon (1.1 eV); glass/PVC with a gap above about 5 eV; free electrons and holes increasing with temperature in silicon; n-type (phosphorus,

Search ARES for similar videos EXERCISE: Electric Conductors and Insulators Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	germanium (0.67 eV), silicon (1.1 eV) and glass/PVC (above about 5 eV); (2) arrows or dots showing more free electrons and holes in silicon at higher temperature; (3) n-type silicon with phosphorus (extra free electrons) and p-type silicon with boron (holes), with a note that doped silicon is still electrically neutral overall; (4) a short caption connecting each diagram to the bulb test (copper bright, glass, wood and plastic off, diode lit forward-biased and off reverse-biased). Each group marks with a coloured pen one change they made to the model because of the simulation evidence. Groups then do a 'gallery walk', placing a tick or a question mark on another group's model. Each learner completes the summary table.	because of evidence. Write the reason next to it." Circulate and ask: "Where in your model can I see the energy gap? Where are the free electrons? Where are the holes?" WAIT TIME of 10 to 15 seconds. For the gallery walk (4 minutes), say: "Leave one tick for something clear and one question mark for something unclear." Cold-call 3 groups to explain one change and the evidence for it. Close by saying: "Our model now has digital evidence behind it. In Lesson 6 we will test it on a new question: what happens to resistance when a material is cooled below a critical temperature, and how does that help an MRI machine?" Collect the summary tables.	prepares it for the superconductor and MRI evidence.	majority carriers electrons) and p-type (boron, majority carriers holes) shown clearly with a note on overall neutrality; captions linking to the bulb test. Indicator of progress: at least one coloured change per group with a written reason that cites simulation evidence. Use the gallery walk question marks to identify the most common unclear ideas to address at the start of Lesson 6.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did every learner change only one variable at a time, and did the checker role help? (2) Which predictions were most often not supported, and did learners treat that as useful evidence rather than a mistake? (3) Did anyone say insulators have zero conduction or that doped silicon is charged, and was this corrected using the simulation evidence? (4) Were learners able to name at least one limit of the simulation compared with the real bulb test? (5) Did the group roles give quieter learners a chance to contribute, and were devices shared fairly? (6) Which new DQB questions point to Lesson 6 (superconductors and MRI) and Lesson 7 (solar panels, microchips and the Garissa solar plant)? Note what to revisit at the start of Lesson 6 based on the gallery walk question marks.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did the simulation show when we changed the energy gap (metal 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulator above about 5 eV), raised the temperature of silicon, and added phosphorus or boron? Record the numbers and the number of free electrons and holes you saw.
What did I learn?	What did the simulation evidence tell us about how well our band model works? Which parts did it support, which parts did we have to change, and what are the limits of a simulation compared with the real bulb test in the Nairobi wiring circuit?
How does this explain the phenomenon?	Using the evidence, explain why copper lights the bulb while glass, wood and plastic almost do not, why silicon sits in between and conducts more when warmer, and how phosphorus (n-type) and boron (p-type) change it. Link your answer to why Mr Otieno can safely hold PVC-covered copper cable.

LESSON 6 (80 minutes (2 x 40-minute periods)): Superconductors and MRI Magnets: When Resistance Drops to Zero

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students extend the class model of resistance to current by investigating superconductors. They find that below the critical temperature a superconductor has zero resistance, and they use this to explain how MRI magnets work in hospitals such as Kenyatta National Hospital.
Knowledge	A superconductor is a material whose resistance drops to zero below its critical temperature (T_c). Mercury has $T_c = 4.2\text{ K}$ (-269 C), niobium-titanium (NbTi) has $T_c = 9.2\text{ K}$ and YBCO has T_c about 92 K (-181 C). Liquid nitrogen boils at 77 K (-196 C), which is below the T_c of YBCO, and liquid helium at 4.2 K cools NbTi. A YBCO disc cooled to 77 K levitates a magnet (Meissner effect). NbTi magnets in MRI machines carry large current without energy loss and give fields of about 1.5 T to 3 T . Copper, by comparison, has resistivity of 1.7×10^{-8} ohm metre at about 20 C and its resistance rises slightly with temperature.
Skills	Developing and using models (revising the class model to show zero resistance); analysing and interpreting data (comparing T_c values with the 77 K and 4.2 K coolants); constructing explanations from evidence; obtaining, evaluating and communicating information (video or demonstration, Dr Wekesa's MRI case).
Attitudes	Curiosity about extreme-temperature materials; appreciation of how physics supports healthcare in Kenya; respect for others' ideas during discussion; responsibility when handling cryogenic liquids or watching the demonstration safely.
Key Inquiry Question	How do semiconductors and superconductors differ from ordinary conductors and insulators? How is electronics applied in day-to-day life?
Purpose in Storyline	Lessons 1 to 5 explained why copper conducts, why plastic, glass and wood almost do not, and how semiconductors sit in between and can be doped. This lesson takes resistance to its extreme: some materials have none at all. It answers the part of the Driving Question about MRI machines and prepares students for the Lesson 7 case studies and final model-based answer.
Safety Notes	Liquid nitrogen at 77 K (-196 C) causes severe cold burns and frostbite. ONLY the teacher (Ms Achieng) handles it, wearing safety goggles, insulated gloves and closed shoes, in a well-ventilated room, with students at least 2 m away. Never seal liquid nitrogen in a closed container. Use plastic tweezers for the YBCO disc and keep magnets away from phones and bank cards. If no liquid nitrogen is available, use the video demonstration instead. Students with pacemakers or implants should keep away from strong magnets. Keep the 3 V test circuit away from water.

B. LESSON OVERVIEW

Back in Nairobi, Mr Otieno's copper cable carries current well, but even copper has some resistance, so some energy becomes heat. Wanjiku asks: 'Can any material have no resistance at all?' Students first predict what happens to the resistance of a metal as it gets very cold. Ms Achieng then runs the liquid nitrogen demonstration (or shows a video) of a YBCO disc cooled to 77 K levitating a magnet. Students compare the T_c values of mercury, NbTi and YBCO with the 77 K and 4.2 K coolants and explain Dr Wekesa's MRI magnet at Kenyatta National Hospital. They add zero resistance to the class model, add new questions to the Driving Question Board (DQB), and complete the summary table.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: All about superconductors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electronics, superconductor, critical temperature	Students recall the Lesson 1 test circuit, in which copper made the bulb bright while glass, wood and plastic left it off, and the Lesson 2 resistivity of copper (1.7×10^{-8} ohm metre). Wanjiku asks if a material could have no resistance. Individually, students draw a sketch graph of resistance against temperature for copper from 300 K down to 0 K and write one sentence predicting what happens to the resistance of a very cold metal. They then predict what will happen when a magnet is placed above a YBCO disc cooled in liquid nitrogen. They share in pairs.	Display the question: 'Mr Otieno's copper cable carries current, but does it carry it with zero resistance? What do you think happens if we cool a metal to very low temperatures?' Say: 'Sketch your own graph first. There is no wrong prediction today.' Give 2 minutes of silent drawing, then WAIT TIME of 10-15 seconds after asking for one volunteer. Cold-call 4 students (2 whose graph slopes to a small value, 2 whose graph drops to zero) and record predictions on the board without judging. Ask: 'What does resistance mean in our class model of electrons moving through copper?' and cold-call 2 more students.	Predict-Observe-Explain with a sketch graph: committing to a prediction in drawing exposes prior ideas (such as 'resistance just gets smaller slowly'), so students have a stake in the evidence they will see in the Observe Phase.	Check the sketches: Does the student draw a graph with resistance falling with temperature (partial understanding), staying constant, or dropping abruptly to zero? Can the student link resistance to electrons colliding with atoms in the Lesson 1 model? Record the count of each prediction type on the board for comparison after the demonstration.
Observe Phase  VIDEO: All about superconductors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electronics, superconductor, critical temperature	Students watch Ms Achieng's demonstration (or the video) of a YBCO disc at room temperature, with a small magnet that does not levitate. The disc is then cooled in liquid nitrogen at 77 K (-196 C) and the magnet floats above it (Meissner effect). Groups record observations in a table: temperature (K), whether the magnet levitates, and what happens as the disc warms above its T_c of about 92 K. Groups are then given a data card with: mercury $T_c = 4.2$ K (-269 C); NbTi $T_c = 9.2$ K; YBCO T_c about 92 K (-181 C); liquid nitrogen = 77 K; liquid helium = 4.2 K; copper resistivity 1.7×10^{-8} ohm metre at about 20 C. They convert temperatures using $K = C + 273$ and mark which coolant works for each material. Finally, they watch a short clip or diagram of Dr Wekesa's MRI machine at a Nairobi hospital with a 1.5 T to 3 T	Say: 'Everyone stand back at least 2 m. I am the only one handling the liquid nitrogen, because it is -196 C.' Pour with goggles and gloves and narrate: 'Watch the magnet. Write down exactly what you see, not what you think should happen.' After levitation, ask: 'What is different about the disc now compared to room temperature?' WAIT TIME 10-15 seconds, then cold-call 3 students. Circulate while groups complete the data card, asking: 'Is 77 K above or below the T_c of YBCO? Is 77 K below the T_c of NbTi? So why does NbTi need liquid helium at 4.2 K?' Cold-call 3 groups to share their coolant matches. If no liquid nitrogen is available, play the video and pause at the levitation moment.	Structured data table with evidence-first observation: students separate observation (magnet floats below T_c, not above) from inference (resistance is zero), and converting units forces them to compare real numbers rather than rely on impressions.	Observable indicators: students record 'magnet floats only when the disc is below its T_c'; correctly convert -196 C to 77 K and -269 C to 4 K; state that 77 K is below the T_c of YBCO (about 92 K) but above the T_c of NbTi (9.2 K) and mercury (4.2 K). Note groups that wrongly say liquid nitrogen can cool NbTi into superconductivity and address this immediately.

	magnet.			
Explain Phase  VIDEO: All about superconductors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electronics, superconductor, critical temperature	Students revise their Predict graphs into a corrected graph: resistance falls gradually with temperature as for copper, then for a superconductor drops abruptly to zero at T_c. In groups they build an explanation of what zero resistance means for current: with zero resistance, current in a loop can flow without energy being lost as heat. They then answer: 'How does Dr Wekesa's MRI machine use NbTi?' using the data: a large current in NbTi coils creates a 1.5 T to 3 T field without energy loss, but only if the coils are kept at 4.2 K with liquid helium, below the T_c of 9.2 K. Using the Lesson 2 resistivity idea, they compare copper (1.7×10^{-8} ohm metre, resistance that rises slightly with temperature) with a superconductor below T_c (zero). They write a Claim-Evidence-Reasoning (CER) statement: 'A superconductor has zero resistance below T_c.' Students also discuss why household wiring still uses copper and PVC rather than superconductors.	Say: 'Compare your Predict sketch to what we saw. What changed in your thinking?' Cold-call 4 students; WAIT TIME 10-15 seconds before each. Ask: 'If the resistance is zero, where does the energy that would have become heat go?' and 'Why would Dr Wekesa's MRI magnet at Kenyatta National Hospital be very wasteful if it were made of copper?' Push for evidence: 'Which number on your data card supports that?' Ask: 'Why does Mr Otieno still use copper and PVC in the house?' Guide students to say that room temperature is far above every T_c listed (mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K), so cooling would be needed. Correct: 'Superconductors are not the same as conductors; a superconductor has zero resistance only below T_c.' Cold-call 3 groups to read their CER aloud.	Claim-Evidence-Reasoning with graph revision: students defend a claim with demonstration and data card evidence and connect it to the model of electrons and resistance from Lesson 1, so the explanation rests on the same mechanism rather than new facts.	Look for CER statements that: (claim) a superconductor has zero resistance below T_c; (evidence) the YBCO disc at 77 K levitates a magnet, 77 K is below its T_c of about 92 K; (reasoning) zero resistance allows large current with no energy lost as heat, so NbTi at 4.2 K gives 1.5 T to 3 T in MRI. Check that students use the terms conductor, superconductor and critical temperature correctly and do not call a superconductor an insulator.
Driving Question Board (DQB) Creation  VIDEO: All about superconductors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading</i>	Students revisit the class DQB kept by Wanjiku and Kamau. Using sticky notes, they mark questions answered by this lesson (e.g. 'Can any material have no resistance?'; 'How do MRI machines work?') and write new ones such as 'Why does YBCO levitate a magnet?' and 'Could we ever have a superconductor at room temperature for Kenyan power lines?' They sort the notes into 'Answered', 'Partly answered' and 'New'.	Say: 'Go to the DQB. Which questions can we now answer with evidence from today? Which are new?' Give 3 minutes for writing, then WAIT TIME 10-15 seconds and cold-call 4 students to read one note and say which evidence supports their move. Ask: 'Which question will Lesson 7 help us answer, linking solar panels, microchips and MRI?' Point out the connection to the Driving Question about household wiring, solar panels, computers and MRI machines.	Driving Question Board sorting: tracking questions as answered, partly answered or new makes visible how evidence builds toward the Driving Question and keeps students' own curiosity driving the unit.	Check that at least 8 students add a new note, that questions are specific (mentioning T_c, MRI or YBCO), and that moved notes are justified with evidence from the demonstration or data card. Note students who ask about practical use, as they are ready for the Lesson 7 case studies.

in the ARES library for this activity. Search ARES for readings Search terms: electronics superconductor critical temperature				
Model Building Phase  VIDEO: All about superconductors Source: Khan Academy (English - US curriculum) Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: electronics superconductor critical temperature	Groups revise the class model of electrons, atoms and energy bands. They add a new panel for a superconductor below T_c: electrons move through the material with no resistance, shown as a smooth arrow with no collisions and no heat, beside the copper panel (electrons collide with atoms, slight resistance). They add temperature labels for T_c (mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K) and a labelled sketch of an NbTi coil in an MRI machine at 4.2 K with liquid helium and a 1.5 T to 3 T magnetic field. Finally they complete the summary table (What we observed / What we learned / How we explain it).	Say: 'Add a superconductor panel to our class model. Show what is different from copper and from silicon. Use arrows and labels, and show the temperature.' Circulate and ask: 'Where on your model is the resistance? Why is the arrow smooth here?' and 'What would Dr Wekesa say happens if the NbTi warms above 9.2 K?' (resistance returns, so the magnet stops working). WAIT TIME 10-15 seconds, then cold-call 3 groups to explain one feature. Remind them that the model still covers conductors (copper), insulators (glass, wood, plastic) and semiconductors (silicon, germanium, diode). Collect the summary tables as an exit check.	Cumulative model revision: adding one new panel to the shared class model shows how zero resistance fits with the existing ideas, helping students see superconductors as a different case of the same electrons-and-resistance story.	Observable indicators: the model shows zero resistance below T_c with no energy lost as heat; T_c values and coolants (77 K nitrogen, 4.2 K helium) are labelled correctly; the MRI sketch links a large current in NbTi to a 1.5 T to 3 T field. Exit question: 'Why must the NbTi in Dr Wekesa's MRI machine stay at 4.2 K?' Look for the answer: it must stay below its T_c of 9.2 K to keep zero resistance.

D. TEACHER REFLECTION

After the lesson, ask yourself: Did students separate what they saw (magnet floating below T_c) from what they inferred (zero resistance)? Did anyone confuse superconductors with ordinary conductors or think liquid nitrogen could make every material superconducting? Did temperature conversions ($K = C + 273$) slow groups down? Which students used evidence from the data card rather than recalled facts? If liquid nitrogen was unavailable and you used video, did students still engage with the levitation moment? What would you change about the questions on the DQB before Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a YBCO disc at room temperature did not levitate a magnet, but cooled to 77 K (-196 C) in liquid nitrogen it made the magnet float. We compared T _c values: mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K, and noted which coolant (77 K or 4.2 K) is cold enough for each.
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What did I learn?	We learned that a superconductor has zero resistance below its critical temperature (T_c), unlike copper, which always has some resistance (1.7×10^{-8} ohm metre at about 20 C). With zero resistance a large current flows with no energy lost as heat.
How does this explain the phenomenon?	We explain that Dr Wekesa's MRI machine uses NbTi coils cooled to 4.2 K with liquid helium, below the T_c of 9.2 K, so the magnet carries a large current without energy loss and gives a field of about 1.5 T to 3 T. Household wiring still uses copper and PVC because room temperature is far above every T_c we studied.

LESSON 7 (80 minutes (double lesson)): Case Studies and the Final Answer to the Driving Question

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To apply all the ideas from Lessons 1 to 6 to three Kenyan technology cases (solar panels, microchips and superconducting MRI magnets), to compare the Lesson 1 class model with the final class model, and to write a final model-based answer to the Driving Question.
Knowledge	Learners explain that copper conducts because its outer electrons are free to move (bands overlap, 0 eV gap), while glass, wood and plastic have a large energy gap (above about 5 eV) so almost no electrons can move, making PVC a safe cover for wires. Silicon (1.1 eV) and germanium (0.67 eV) have a small gap, conduct a little, and conduct more when warmer. Doping with phosphorus (n-type) or boron (p-type) raises conductivity in a controlled way, and joining p and n gives a diode, the basis of solar panels and microchips. Superconductors below their critical temperature have zero resistance, so NbTi magnets in MRI machines carry large current without energy loss.
Skills	Analysing case studies; using evidence to develop and revise models (NGSS SEP 2); constructing explanations (NGSS SEP 6); engaging in argument from evidence (NGSS SEP 7); obtaining, evaluating and communicating information (NGSS SEP 8).
Attitudes	Appreciate the role of electronics in Kenya's technological growth (solar energy at Garissa, mobile money on phones, medical imaging at Kenyatta National Hospital). Show open-mindedness, respect for peers' ideas and responsibility when critiquing models.
Key Inquiry Question	1. How do semiconductors and superconductors differ from ordinary conductors and insulators? 2. How is electronics applied in day-to-day life?
Purpose in Storyline	This is the final lesson of the storyline. Learners return to Mr Otieno's copper cable with its PVC cover and the five-material test (copper bright, glass, wood and plastic off, diode lit in one direction only). They use Lessons 1 to 6 evidence to finalise the class model, complete the Driving Question Board (DQB) and write the Final Explanation answering why some materials conduct while others do not and how this affects safety and efficiency in wiring, solar panels, computers and MRI machines.
Safety Notes	This lesson is mostly discussion, case study reading and writing. If the 3 V test circuit (two 1.5 V dry cells, 3 V bulb, crocodile clips) is re-run for a quick recall, use only the dry cells, never mains 240 V. If the liquid nitrogen video is shown, make clear that liquid nitrogen (77 K, -196 C) is handled only by trained adults with goggles and gloves. Remind learners never to touch mains wiring, even when it appears insulated.

B. LESSON OVERVIEW

Ms Achieng opens by asking learners to predict how the five-material results would be explained to Mr Otieno's client in Nairobi. Groups then work on one of three Kenyan case studies: the Garissa solar plant (50 MW, silicon p-n junctions), microchips in phones, M-Pesa and computers (silicon with billions of transistors), and the Kenyatta National Hospital MRI (NbTi magnets, zero resistance below T_c , field about 1.5 T to 3 T, Dr Wekesa). Learners use the class model, summary tables and posters from Lessons 1 to 6 as evidence. They compare the Lesson 1 model, which only showed electrons and atoms, with the final model that includes energy bands and gaps, free electrons, holes, doping and critical temperature. They complete the DQB by sorting questions as answered or still open, and each learner writes a Final Explanation that answers the Driving Question in their own words. Throughout, the language is conductor, insulator, semiconductor and superconductor, never mixed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners look at the class poster of the five-material test (copper bright; glass, dry wood and plastic strip off; 1N4007 diode lit forward-connected and off reverse-connected). Individually they write 2 or 3 sentences predicting: 'If you had to explain to Mr Otieno why he can safely hold the PVC-covered copper cable at 240 V, which three ideas from our unit would you use?' They also predict which of the three Kenyan technologies (solar panel at Garissa, microchip in an M-Pesa phone, MRI magnet at Kenyatta National Hospital) uses a conductor, a semiconductor or a superconductor. They share with a partner.	Display the Lesson 1 class model and the five-material results. Say: 'Today we finish our story. Mr Otieno holds a live-looking cable safely. Before we look at any new evidence, write down which ideas you would use to explain that.' Give WAIT TIME of 10-15 seconds before anyone writes. Walk around and note ideas. Cold-call 4 learners (not volunteers) to share one idea each. Then ask: 'Which of our three Kenyan cases uses a semiconductor? Which uses a superconductor? Turn and tell your partner.' Record predictions on the board without judging them. Say: 'We will check these against evidence in a few minutes.'	Predict-Observe-Explain with Think-Pair-Share: writing a prediction first activates prior knowledge from Lessons 1 to 6 and makes learners accountable for their ideas. The shared record lets them later see how their thinking developed.	Look for: (1) learners who mention free electrons, energy gap or bands rather than only 'plastic does not allow electricity'; (2) learners who correctly match semiconductor to the solar panel and microchip and superconductor to the MRI magnet; (3) any learner who says insulators have zero conduction, or who mixes up conductor and semiconductor, to be addressed later. Tally on a class list how many of the 4 cold-called learners used the term energy gap.
Observe Phase  VIDEO: 06 P N Junctions And Diodes Source: Seavuria  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners work in three expert groups, one per Kenyan case. Group A (Garissa solar plant, 50 MW; home solar kits): read the case card on silicon p-n junctions and identify the n-type (phosphorus) and p-type (boron) layers, and the role of the junction. Group B (microchips in phones, M-Pesa and computers): read about silicon chips with billions of transistors and why controlled doping and one-way behaviour (like the 1N4007 diode) matter. Group C (Kenyatta National Hospital MRI, Dr Wekesa): read about NbTi magnets (T _c 9.2 K) cooled by liquid helium at 4.2 K, field about 1.5 T to 3 T, zero resistance below T _c , with a short recap video or the YBCO levitation demonstration at 77 K (Meissner effect). Each group fills a 'Case Evidence Card' with three	Distribute case cards and the Evidence Card template. Say: 'Your job is to find out which kind of material does the key job in your case and which property from our class model explains it.' Circulate and ask: 'Which energy gap matters here?' 'Where in your case do you see the one-way action we saw with the diode?' 'Why does zero resistance mean no energy loss in the MRI magnet?' Allow WAIT TIME of 10-15 seconds after each question. For Group C, remind: 'Superconductor means zero resistance only below T _c . NbTi needs 4.2 K; YBCO works at 77 K.' Cold-call 3 learners from different groups to state one fact from their card before the jigsaw. During the jigsaw, listen for correct vocabulary and gently correct any use of 'half conductor'. Also note that copper cable with PVC is a	Jigsaw with case-based evidence gathering: every learner becomes responsible for one case and must teach it, which ensures all three applications are linked to the same underlying ideas (energy gap, doping, critical temperature). Using a common Evidence Card makes cross-case patterns visible.	Check Evidence Cards for: Group A naming p-type (boron, holes) and n-type (phosphorus, electrons) layers and silicon as semiconductor; Group B stating that doping of about 1 atom in 10 ⁷ to 10 ⁸ controls conductivity and that the p-n junction conducts one way; Group C stating zero resistance below T _c , NbTi at 9.2 K cooled by helium at 4.2 K, and field about 1.5 T to 3 T. During the jigsaw, count how many learners can teach their case without reading from the card. Note learners who use 'free electrons', 'holes', 'forward-biased' correctly.

	columns: material type (conductor, insulator, semiconductor or superconductor), property that matters, and what would go wrong without it. Then groups rotate in a jigsaw so each learner teaches their case to two others.	conductor plus insulator pair, to link back to Mr Otieno.		
Explain Phase  VIDEO: 03 Intrinsic And Extrinsic Semi Conductors Source: Seavuria  Search ARES for similar videos  HTML: Circuit Construction Kit: DC <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	In mixed groups of four, learners build a 'Why does it work?' concept chain linking each case back to the phenomenon. They complete the sentence frames: 'Copper carries current in Mr Otieno's cable because ...', 'PVC makes the cable safe because ...', 'Silicon in a solar panel or chip is useful because ...', 'NbTi in the MRI magnet carries large current without energy loss because ...'. They then use the resistivity order (copper 1.7×10^{-8} ohm metre, germanium 0.46, silicon 2.3×10^{-3} , dry wood about 10^8 to 10^{11} , glass about 10^{10} to 10^{14} , PVC about 10^{13} or more) and energy gaps (metals 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulators above about 5 eV) to justify the ranking. Finally each group prepares a 1-minute argument answering: 'Which one material property explains safety in wiring and efficiency in devices?'	Say: 'Use evidence, not just words. Every claim needs a number or an observation from our unit.' Pose: 'Silicon is about 10^{11} times more resistive than copper, yet it powers Garissa and your phone. How can both be true?' Give WAIT TIME of 10-15 seconds, then ask groups to discuss for 2 minutes. Cold-call 3 groups to share, and ask a different group to agree, disagree or add evidence each time ('Who can build on that?'). Probe: 'What changes when we dope silicon?' 'What does the diode in our test circuit tell us about a solar cell?' 'Why is the PVC cover a good choice and not glass?' If a learner says insulators conduct nothing, respond: 'Say almost none. What does the large gap above about 5 eV do?' Do not re-teach lesson content; refer back to the class posters and summary tables instead.	Claim-Evidence-Reasoning (CER) with peer argumentation: learners write a claim, back it with data from the posters and tables, and give reasoning with the energy band model. Peer challenge exposes gaps and strengthens explanations.	Listen for reasoning that: links 0 eV gap (bands overlap) to free electrons in copper; links gap above about 5 eV to almost no moving electrons in PVC, wood and glass; explains that doping greatly raises silicon conductivity in a controlled way and the p-n junction gives one-way conduction; explains zero resistance below T_c as no energy lost as heat. Use a quick checklist of these four points per group. Flag misconceptions such as 'doped silicon is charged' (it remains electrically neutral overall) or 'superconductors work at room temperature'.
Driving Question Board (DQB) Creation  VIDEO: 05 Doping A Semiconductor Source: Seavuria  Search ARES for similar videos  HTML: Electric	Learners revisit the class DQB, which has been kept by Wanjiku and Kamau's class throughout the unit. In pairs they take the question sticky notes and sort them into three groups: 'Answered with evidence', 'Partly answered', and 'New or still open questions'. For each answered question they attach the evidence source (for example the five-material test, resistivity table, doping explanation, PhET simulation,	Say: 'Our board started with puzzles. Let us see how many we can now answer, and what evidence supports each answer.' Hand out the sticky notes and allow WAIT TIME of 10-15 seconds before pairs begin sorting. Circulate and ask: 'Which lesson gave you the evidence for this answer?' Cold-call 4 pairs to place one question and justify where it belongs. Ask the class: 'Which questions are still open,	DQB sorting with evidence tagging: connecting each answered question to a specific piece of evidence shows how knowledge was built over the unit, and marking open questions models how scientists keep asking.	Check that at least most of the DQB questions are correctly placed and that each 'Answered' note names a valid evidence source. Look for learners who write new questions that extend ideas (for example about cost, temperature, efficiency or safety) rather than repeat old ones. Note pairs who cannot link a question to a lesson for follow-up.

Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	liquid nitrogen demonstration or video). Each learner then adds one new question that could be investigated next, such as 'Can we make solar panels cheaper in Kenya?' or 'Could a superconductor work at room temperature?'	and why does that make science exciting?' Make sure the two KICD key inquiry questions are visible and answered: how semiconductors and superconductors differ from ordinary conductors and insulators, and how electronics is applied in day-to-day life. Photograph the completed DQB for the record.		
Model Building Phase  VIDEO: 06 P N Junctions And Diodes Source: Seavuria Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners compare the Lesson 1 class model (electrons and atoms with the puzzle of why materials differ) to the final class model. Individually they draw a final model on A4 paper or in their exercise book showing: energy bands with the gap for a metal like copper (0 eV, bands overlap), a semiconductor such as silicon (1.1 eV) and germanium (0.67 eV), and an insulator such as glass or PVC (above about 5 eV); n-type (phosphorus, extra free electrons) and p-type (boron, holes) silicon; the p-n junction in a diode; and a note on superconductors below T_c. They add arrows or labels showing where each fits in Kenyan cases: Mr Otieno's PVC-covered copper cable, Garissa solar plant, microchips, and the Kenyatta National Hospital MRI. Then each learner writes their Final Explanation answering the Driving Question: 'Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines?' Use the sentence starters: 'Copper conducts because ...', 'Glass, wood and plastic ...', 'Silicon and germanium ...',	Display the Lesson 1 model beside a blank space for the final one. Say: 'Look at our first model. What could it not explain? What can our final model explain now?' Give WAIT TIME of 10-15 seconds. Cold-call 3 learners to name one thing the final model explains better. Remind learners that doped silicon is electrically neutral and that insulators have almost none, not zero, conduction. While they draw and write, circulate with a checklist and prompt: 'Where is the energy gap in your drawing?' 'Which carriers are the majority in p-type?' 'How does your model explain why Mr Otieno can hold the insulated wire safely?' Share the exemplar outcome statement only after learners have drafted their own: copper conducts because its outer electrons are free to move through the metal; glass, wood and plastic have a large gap above about 5 eV so almost no electrons move, which makes plastic a safe cover; silicon and germanium have small gaps and conduct a little, more when warmer, and doping with phosphorus or boron greatly raises conductivity in a controlled way; joining p and n gives a diode that conducts one way only, the basis of solar panels and microchips;	Model revision and comparison (before and after) combined with a written Final Explanation: contrasting the Lesson 1 model with the final model makes the growth in understanding explicit, and writing the explanation in their own words consolidates the cause-and-effect reasoning from energy gap to function.	Use a rubric with four observable indicators: (1) the model correctly shows gaps in order metals 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulators above about 5 eV; (2) shows n-type with free electrons and p-type with holes, and the one-way p-n junction; (3) mentions zero resistance of superconductors below T_c with NbTi in MRI; (4) the Final Explanation links properties to safety (PVC cover) and efficiency (copper wiring, solar, chips, MRI) using correct vocabulary and Kenyan examples. Record learners who meet at least 3 indicators and plan support for others.

	'Superconductors ...'. Pairs swap and give one 'glow' and one 'grow' comment before final edits.	superconductors below T _c have zero resistance so NbTi magnets in MRI machines carry large current without energy loss. Learners then compare it with their draft and revise.		
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D. TEACHER REFLECTION

After the lesson, ask: Did learners use evidence from earlier lessons (the five-material test, resistivity order, energy gaps, doping, T_c values) instead of memorised statements? Which learners still confuse semiconductors with superconductors or say insulators conduct zero? Were all three Kenyan cases (Garissa solar plant, microchips, Kenyatta National Hospital MRI) understood as applications of the same ideas? How did the final model differ from the Lesson 1 model in each learner's work? Did the Final Explanation answer both parts of the Driving Question, safety and efficiency? Which DQB questions remain open and could become project work or career talks (electrician, radiographer, solar technician, electronics engineer)? Note adjustments for the next class, including more time for model drawing or a repeat of the liquid nitrogen video if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: copper lit the bulb brightly; glass, dry wood and plastic did not; the silicon diode (1N4007) lit the bulb only when forward-connected. In the Kenyan cases we saw silicon p-n junctions at work in the Garissa solar plant (50 MW) and in microchips, and an NbTi superconducting magnet cooled to 4.2 K in the Kenyatta National Hospital MRI, with a YBCO disc levitating a magnet at 77 K.
What did I learn?	What we learned: materials differ in resistivity (copper 1.7×10^{-8} ohm metre up to PVC about 10^{13} or more) and in energy gap (metals 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulators above about 5 eV). Doping silicon with phosphorus (n-type) or boron (p-type) raises conductivity in a controlled way, and a p-n junction conducts one way only. Superconductors have zero resistance below their critical temperature.
How does this explain the phenomenon?	How we explained it: Copper conducts because its outer electrons are free to move (bands overlap), while glass, wood and plastic have a large gap so almost no electrons can move, making PVC a safe cover for Mr Otieno's wires. Silicon and germanium conduct a little, more when warmer, and doping with phosphorus or boron greatly raises conductivity, so p-n junctions make solar panels and microchips work. Below T _c , NbTi has zero resistance so MRI magnets carry large current without energy loss, which answers our Driving Question.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.